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LAN PROPERTY FOR THE PARAMETER OF THE JUMP RATE IN MEAN FIELD INTERACTING SYSTEMS OF NEURONS

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Abstract. In the context of a large system of N neurons interacting through spike events in a mean-field regime as $N \to \infty$, we characterize the estimation of a multidimensional parameter in the spiking rate, when the neural states are observed over a fixed time horizon. We first prove the local asymptotic normality (LAN) property and leverage classical theory to establish the asymptotic efficiency of the maximum likelihood estimator. While the theory of Ibragimov and Has'minskii yields strong results, up to global asymptotic minimax bound, its applicability appears currently limited to models without state resets at spike times. Following then Höpfner's classical approach, we nevertheless derive, in a general setting including neuron reset, the consistency, asymptotic normality and local asymptotic minimax optimality of the estimator.

Keywords: Local Asymptotic Normality (LAN); Mean-field regime; Interacting particle system; Multidimensional parameter estimation; Jump rate estimation; Maximum likelihood estimator (MLE); Asymptotic minimax optimality

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1. Introduction

In modern neuro-mathematics, a substantial body of work focuses on large-scale limits of systems of interacting neurons, particularly in brain regions where neurons exhibit similar behavior and can be modeled through mean-field type interactions (see [9], [10], [12], [13], [14], [16], [21], [34], [33] and the references therein). In this framework, the large population limit yields a mesoscopic description of an individual neuron via a limiting theoretical dynamics, referred to as the mean-field limit, which is typically more tractable than the corresponding large finite system.

In the present work, we investigate the insights that the mean-field limit can provide about the actual finite neuronal system from a statistical standpoint. More specifically, we focus on understanding how a given neuron responds to incoming stimuli through its spiking rate function, which characterizes its ability to react by generating sequences of spikes.

As is standard in the literature, we model the spiking activity using a variant of nonlinear Hawkes processes (see [7], [8], [18], [22], as well as [23] and references therein). More precisely, the model consists of N interacting components (neurons) that evolve according to a deterministic flow between successive spikes of the system and emit spikes at rates depending on their membrane potentials. This dynamics is described by the following system of stochastic differential equations driven by the system (π^j), $j \in \mathbb{N}^*$, of i.i.d. Poisson Random Measures on $\mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}$ with intensity $\mathrm{d}s\,\mathrm{d}z\,\nu(\mathrm{d}u)$, where

ν is a probability measure on $\mathbb{R}$ subject to moment conditions:

$$\mathrm{d}X_t^{i,N,\theta} = b(X_t^{i,N,\theta})\mathrm{d}t + \frac{1}{N}\sum_{j\neq i}^{N}\int_{\mathbb{R}_+}\int_{\mathbb{R}} u\mathbf{1}_{\left\{z\leq f_\theta(X_{s-}^{j,N,\theta})\right\}}\pi^j(\mathrm{d}t,\mathrm{d}z,\mathrm{d}u) + \int_{\mathbb{R}_+}\int_{\mathbb{R}}\varphi(X_{s-}^{i,N,\theta})\,\mathbf{1}_{\left\{z\leq f_\theta(X_{s-}^{i,N,\theta})\right\}}\pi^i(\mathrm{d}t,\mathrm{d}z,\mathrm{d}u)\,, \qquad X_0^{i,N,\theta}= X_0^i\,.$$

In this system $X_i^N(t)$ represents the membrane potential of neuron i in a network of N neurons; a typical description of the flow b is $b(x) = -\alpha x$ where $\alpha > 0$ describes a leak of the potential; the spikes of neuron i are generated by the Poisson measure π^i with conditional intensity $f(X_i^N(t-))$. The function φ describes the instantaneous loss of the potential of neuron i when it emits a spike, and a typical example is $\varphi(x) = -x$, which models the situation where when a neuron i emits a spike, its potential is reset (set to zero). The spikes of neuron j produce jumps of size U/N in any neuron $i \neq j$, where the random height U is generated following the distribution ν, independently of all the past.

Our goal is to estimate the unknown spiking rate function for a fixed membrane potential level, based on continuous-time observations of the neurons' membrane potentials over a fixed time interval $[0, T]$, while the number of observed neurons increases. Thus, the asymptotic regime we consider corresponds to $N \to \infty$ with the observation time window kept fixed.

This work lies within the domain of statistical inference for particle systems and related McKean–Vlasov limits, a very active and rapidly developing research area in recent years (see [25], [3], [4], [1], [2], [15], [11], among others). In most of these papers, Brownian or fractional Brownian noise is considered. Consequently, systems driven by jump processes represent an important and relatively less explored direction within this line of research.

In the recent paper [17], we introduced a kernel estimator of Nadaraya-Watson type and discussed its asymptotic properties with the help of the deterministic dynamical system describing the mean field limit in the absence of the state resets at spike times, i.e. when $\varphi \equiv 0$. We computed the minimax rate of convergence in an L^2-error loss over a range of Hölder classes and obtained the classical rate of convergence $N^{\frac{-2\beta}{2\beta+1}}$, where β is the regularity of the unknown spiking rate function.

In this paper, we assume that the spiking rate function depends on some multidimensional parameter and show the Local Asymptotic Normality (LAN) property as the size of the system $N \to \infty$ with the rate $\sqrt{N}$ and the limiting Fisher information depending on the limit of the empirical measure of the system. We then study the asymptotic properties of the Maximum Likelihood Estimator (MLE). Under the LAN property, we leverage classical theory to establish the asymptotic efficiency of the maximum likelihood estimator. While the approaches of [15] and of Ibragimov and Has'minskii [29] yield strong results, up to global asymptotic minimax bound, its applicability appears currently limited to models without state resets at spike times, i.e. $\varphi \equiv 0$. Following Höpfner's classical approach [27], we nevertheless derive, in a general setting including neuron reset, the consistency, asymptotic normality and local asymptotic minimax optimality of the estimator.

The remainder of this paper is organized as follows. In Section 2, we introduce the mathematical framework for the finite system of interacting neurons (Section 2.1) and its mean-field counterpart (Section 2.2). Our main theoretical results, namely the LAN property and the asymptotic properties of the MLE, are stated in Section 2.3. Section 3 provides a numerical study to clarify the scope of these theoretical results for finite systems, including simulations that illustrate the behavior of LAN quantities and the alignment of the MLE with the theoretical predictions derived from LAN.

The proofs of our results are developed in Sections 4–6: auxiliary lemmas are gathered in Section 4, the proof of the LAN property is presented in Section 5, and the analysis of the MLE (consistency, asymptotic normality, and optimality) is carried out in Section 6. Finally, Appendix A provides technical implementation details and further extends the numerical exploration with additional scenarios (varying system size and an alternative parameter set).

2. Model, assumptions and main results

2.1. **Finite system.** Let $(\pi^j),\ j \in \mathbb{N}^*$ be a family of i.i.d. Poisson Random Measures on $\mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}$ with the intensity $\mathrm{d}s\,\mathrm{d}z\,\nu(\mathrm{d}u)$, where ν a probability measure on $\mathbb{R}$. We consider the following model of interacting spiking neurons:

$$\mathrm{d}X_t^{i,N,\theta} = b(X_t^{i,N,\theta})\mathrm{d}t + \frac{1}{N}\sum_{j\neq i}^{N}\int_{\mathbb{R}_+}\int_{\mathbb{R}} u\mathbf{1}_{\left\{z\leq f_\theta(X_{s-}^{j,N,\theta})\right\}}\pi^j(\mathrm{d}t,\mathrm{d}z,\mathrm{d}u) + \int_{\mathbb{R}_+}\int_{\mathbb{R}}\varphi(X_{s-}^{i,N,\theta})\,\mathbf{1}_{\left\{z\leq f_\theta(X_{s-}^{i,N,\theta})\right\}}\pi^i(\mathrm{d}t,\mathrm{d}z,\mathrm{d}u)\,, \qquad X_0^{i,N,\theta} = X_0^i. \tag{1}$$

The jumping rate f depends on a parameter $\theta \in \Theta$, with Θ a compact subset of $\mathbb{R}^d$. We denote by $\mu^{N,\theta}$ the empirical measure of the system (1), and by $\mu_s^{N,\theta}$ its projection onto the s-th coordinate.

$$\mu_s^{N,\theta}(\mathrm{d}x) = \frac{1}{N}\sum_{j=1}^{N}\delta_{X_s^{j,N,\theta}}(\mathrm{d}x)\,. \tag{2}$$

We impose the following assumptions on the system (1)

Assumption 1. *The parameter set Θ is a compact subset of $\mathbb{R}^d$ with non-empty interior $\mathring{\Theta}$.*

Assumption 2. *(i) The function $b : \mathbb{R} \to \mathbb{R}$ is Lipschitz-continuous,*
(ii) The function $\varphi : \mathbb{R} \to \mathbb{R}$ is Lipschitz-continuous and bounded,
(iii) For all $\theta \in \Theta$ the function : $x \in \mathbb{R} \mapsto f_\theta(x) \in \mathbb{R}_+$ is Lipschitz-continuous and bounded,
(iv) The measure ν satisfies: $\int_{\mathbb{R}} |u|\nu(du) < \infty$ and $\nu(\{0\}) = 0$,
(v) The initial condition is i.i.d. distributed according to a square integrable distribution μ_0.

Under the above assumption, for each fixed N, equation (1) admits a unique strong solution (see [5]).

Assumption 3. *The measure ν satisfies: $\int_{\mathbb{R}} |u|^2\nu(du) < \infty$ and $\nu(\{0\}) = 0$.*

2.2. **Limit system.** As the system size N tends to infinity, under Assumptions (2) and (3), system (1) exhibits the propagation of chaos property [5]: the system (1), extended by setting $\mathrm{d}X_t^{i,N,\theta} \equiv 0, \quad i > N$, converges in law in the infinite product Skorokhod space $\mathcal{D}^{\otimes\infty}(\mathbb{R}_+,\mathbb{R})$ to the infinite collection of i.i.d. processes solving the following McKean-Vlasov SDE:

$$\mathrm{d}\bar{X}_t^{i,\theta} = b(\bar{X}_t^{i,\theta})\mathrm{d}t + m\,\bar{\mu}_t^\theta[f_\theta]\mathrm{d}t + \int_{\mathbb{R}_+}\int_{\mathbb{R}}\varphi(\bar{X}_{s-}^{i,\theta})\,\mathbf{1}_{\left\{z\leq f_\theta(\bar{X}_{s-}^{i,\theta})\right\}}\pi^i(\mathrm{d}t,\mathrm{d}z,\mathrm{d}u)\,, \qquad \bar{X}_0^{i,\theta} = X_0^i\,. \tag{3}$$

Here we denoted $m = \int_{\mathbb{R}} u\ \nu(du)$ and $\bar{\mu}^\theta$ the law on the Skorokhod space of $(\bar{X}^\theta)_{t\geq 0}$, whereas $\bar{\mu}_s^\theta$ denotes its projection on the s-coordinate. Moreover, under Assumptions (2) the system (3) admits a unique strong solution (see [5]).

Remark 2.1. *In the case $\phi(x) = -x$, assuming that the initial condition admits exponential moments, the existence and the uniqueness of solutions to systems* (1) *and* (3)*, as well as the propagation of chaos property, were established in [19]. All our results remain valid in this setting.*

2.3. **Main results.** We fix $t > 0$ and continuously observe on $[0,t]$ the trajectories of all the coordinates X_s^{i,N,θ^*}; $i \leq N$, of the particle system (1) generated under the parameter value $\theta^* \in \mathring{\Theta}$. We denote by Ω_N the N product Skorokhod space $\mathcal{D}^N([0,t],\mathbb{R})$ equipped with the natural filtration $F_N := (\mathcal{F}_s^N)_{s\in[0,t]}$ induced by the canonical mappings

$$X_s^{(N)}(\omega) = (X_s^{1,N}(\omega), \dots, X_s^{N,N}(\omega)) = \omega_s.$$

We write $X^{(N)} := (X_s^{(i,N)})_{i\leq N}$; $s \in [0,t]$, for a generic element of Ω^N and we denote $\P_N^\theta$ the law on (Ω_N, F_N) of the strong solution of (1) and $\mathbb{E}_{\mathbb{P}_N^{\theta^*}}$ the expectation under this law. We study the sequence of statistical experiments

$$(\mathcal{E}^N)_{N\geq 1} = \big(\Omega_N, F_N, (\mathbb{P}_N^\theta, \theta \in \Theta)\big)_{N\geq 1},$$

corresponding to the observation on $[0,t]$ of the solution of (1). Our first main result establishes that, for any $\theta^* \in \mathring{\Theta}$, the Local Asymptotic Normality (LAN) property holds for the sequence of experiments $(\mathcal{E}^N)$ as $N \to \infty$. To formulate this first result, we need to impose additional assumptions.

Assumption 4. *(i) The function $\mathbb{R} \ni x \mapsto f_\theta(x) \in \mathbb{R}_+$ satisfies uniform upper- and lower-bounds, and its Lispchitz constant is bounded:*
$\sup_{\theta\in\Theta}\sup_{x\in\mathbb{R}} f_\theta(x) < \infty$, $\inf_{\theta\in\Theta}\inf_{x\in\mathbb{R}} f_\theta(x) > 0$, $\sup_{\theta\in\Theta}\|f_\theta\|_{Lip} < \infty$.

(ii) For all $x \in \mathbb{R}$, the function $\theta \mapsto f_\theta(x)$ is differentiable, and the gradient, denoted $\dot f_\theta(x) \in \mathbb{R}^d$, satisfies a uniform bound: $\sup_{\theta\in\Theta}\sup_{x\in\mathbb{R}}\|\dot f_\theta(x)\| < \infty$.

Under Assumption 4, we denote

$$\ell_0 = \big(\inf_{\theta,x} f_\theta(x)\big)^{-1},\ B_0 = \sup_{\theta,x}|f_\theta(x)|,\ B_1 = \sup_{\theta,x}\Big\|\frac{\dot f_\theta}{f_\theta}(x)\Big\|, B_L = \sup_\theta \|f_\theta\|_{Lip}, \tag{4}$$

which are all finite and well-defined quantities.

Assumption 5. *There exists $C > 0$ such that for all $\theta, \theta' \in \Theta$,*

$$\left|\frac{f_\theta}{f_{\theta'}}(y) - 1 - (\theta-\theta')^\intercal \cdot \frac{\dot f_{\theta'}}{f_{\theta'}}(y)\right| \leq C\,\|\theta-\theta'\|^2.$$

For any $\theta \in \Theta$, $\theta^* \in \Theta$, the likelihood ratio process $L_t^{\theta/\theta^*} := \dfrac{d\mathbb{P}_N^\theta}{d\mathbb{P}_N^{\theta^*}}$ for the generic trajectory $X^{(N)} \in \Omega_N$ is given by

$$L_t^{\theta/\theta^*}\big(X^{(N)}\big) = \prod_{i=1}^N \prod_{n\geq 1: T_n^i \leq t} \frac{f_\theta}{f_{\theta^*}}\big(X_{T_n^i-}^{(i,N)}\big) \cdot \exp\Big[-\sum_{j=1}^N \int_0^t (f_\theta - f_{\theta^*})\big(X_s^{(j,N)}\big)\mathrm{d}s\Big], \tag{5}$$

where $(T_n^i)_{n\geq 1}$ for $i \in [\![1, N]\!]$ is the ordered sequence of jump times of $X^{(i,N)}$. We omit in the following the dependency on the trajectory $X^{(N)}$.

For all $\theta^* \in \Theta$, denote $I_t^{\theta^*}$ the limiting Fisher information matrix of the sequence of the experiments $\mathcal{E}^N$. It is defined by

$$I_t^{\theta^*} = \int_0^t \bar\mu_s^{\theta^*}\Big[\frac{(\dot f_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\Big]\mathrm{d}s = \int_0^t \bar\mu_s^{\theta^*}\Big[\frac{\dot f_{\theta^*}\cdot[\dot f_{\theta^*}]^\intercal}{f_{\theta^*}}\Big]\mathrm{d}s\,, \tag{6}$$

where $\cdot^{\otimes 2}$ denotes the outer product (tensor product) of the gradient vector.

Denote by $\mathcal{M}_+^d$ the set of symmetric and strictly positive definite matrices in $\mathbb{R}^{d\times d}$.

Assumption 6. *For all $\theta^* \in \Theta$, $I_t^{\theta^*} \in \mathcal{M}_+^d$.*

Remark 2.2. *Assumptions 1, 4 and 6 imply that the limiting Fischer information is also uniformly non-degenerated:*

$$\inf_{\theta^* \in \Theta} \det I_t^{\theta^*} > 0.$$

Remark 2.3. *An example of the rate function that is pertinent in the neuroscience and obviously satisfies Assumptions 4, 5 and 6 is*

$$f_\theta(x) = \theta_1 \arctan x + \theta_2,\ x \geq 0,\ \theta = (\theta_1, \theta_2),$$

$\theta_1 \in \Theta_1 \subset [0,1]$ and $\theta_2 > \pi/2$, $\theta_2 \in \Theta_2$; Θ_1, Θ_2-compacts in $\mathbb{R}_+$.

Another example, where θ_0 corresponds to the maximum intensity, θ_1 to the threshold value of potential, where f_θ has its inflexion point, and θ_2 to the roughness of the transition of f_θ at θ_1

$$f_\theta(x) = \frac{\theta_0}{1 + \exp(-\theta_2 \cdot (x - \theta_1))}, \quad \theta = (\theta_0, \theta_1, \theta_2),$$

is extensively studied in Section 3.

Theorem 2.4. *Under Assumptions (1) to (6) the LAN property for the sequence of experiments $(\mathcal{E}^N)_{N \geq 1}$ holds for any $\theta^* \in \Theta$ at rate $1/\sqrt{N}$ and with the Fisher information matrix $I_t^{\theta^*}$. Namely, there exist for any $t > 0$ a random sequence $(\Delta_t^{N,\theta^*})_N$, of vectors with $\Delta_t^{N,\theta^*} \in \mathbb{R}^d$, and a random sequence $(I_t^{N,\theta^*})_N$, of matrices with $I_t^{N,\theta^*} \in \mathcal{M}_+^d$, such that for all $h \in \mathbb{R}^d$:*

$$\log \frac{L_t^{\theta^* + h/\sqrt{N}}}{L_t^{\theta^*}} = h^\mathsf{T} \Delta_t^{N,\theta^*} - \frac{1}{2} h^\mathsf{T} I_t^{N,\theta^*} h + o_{\mathbb{P}_N^{\theta^*}}(1), \tag{7}$$

and the following convergences hold when $N \to \infty$:

$$\Delta_t^{N,\theta^*} \longrightarrow \mathcal{N}(0, I_t^{\theta^*}), \ \text{ in law under } \mathbb{P}_N^{\theta^*};$$

and

$$I_t^{N,\theta^*} \longrightarrow I_t^{\theta^*}, \ \text{ in } \mathbb{P}_N^{\theta^*} - \text{probability};$$

where $\mathcal{N}(0, \Sigma)$ denotes the d-dimensional Gaussian distribution with mean zero and covariance matrix Σ.

LAN property implies that locally around a given point θ^*, the experiment $\mathcal{E}^N$ parametrized by $\theta = \theta^* + N^{-1/2} h$ behaves asymptotically like the simplest possible experiment, namely a Gaussian shift experiment. This property has powerful consequences for the optimality of estimators through the celebrated Hajek convolution theorem [24], which states that under LAN the limiting law of any regular estimator is the convolution of the normal law with variance $(I^{\theta^*})_t^{-1}$ and an independent error distribution. In particular, the following local asymptotic minimax lower bound holds for arbitrary loss function $w : \mathbb{R}^p \to [0, \infty)$ which are continuous, bounded and subconvex:

Corollary 2.5. *For any sequence of estimators $(\tilde{\theta}_N)$ which estimation errors in θ^* are tight at rate $\sqrt{N}$ and for any continuous, bounded and subconvex loss function $w : \mathbb{R}^p \to [0, \infty)$, it holds that*

$$\liminf_{C \to \infty} \liminf_{N \to \infty} \sup_{\sqrt{N}|\theta' - \theta^*| \leq C} \mathbb{E}_{\mathbb{P}_N^{\theta'}} \Big[w\Big(N^{1/2} (I_t^{\theta^*})^{1/2} \big(\tilde{\theta}_t^N - \theta' \big) \Big) \Big] \geq \frac{1}{(\sqrt{2\pi})^d} \int_{\mathbb{R}^d} w(x)\, e^{-\frac{1}{2}\|x\|^2} \mathrm{d}x\,.$$

This result is proved in [27], (7.12). We exploit now the LAN property to deduce the results of asymptotic minimax optimality for the Maximum Likelihood Estimator (MLE), showing in particular that MLE achieves the lower bound of Corollary 2.5, see Theorem 2.8 bellow. Assume that the trajectory $X^{(N)}$ is generated under an unknown parameter θ^*. The maximum likelihood estimator $\hat{\theta}_t^N$ of θ^* is given by

$$\hat{\theta}_t^N = \mathrm{argmax}_{\theta\in\Theta} L_t^{\theta/\theta^*}(X^{(N)}).$$

To formulate our results about the MLE we need to impose additional assumptions.

Recall the definition of the Fisher information $I_t^{\theta^*}$ in (6). The assumption that $I_t^\theta \in \mathcal{M}_+^d$ holds for any θ entails the local identifiability of the parameter. We need moreover to impose an identifiability assumption implying that the mapping $\theta \mapsto \mathbb{P}_N^\theta$ is one-to-one, which we express in terms of

$$\mathcal{I}_t^{\theta/\theta'} = \int_0^t \bar{\mu}_s^{\theta'}\Big[\Big|\frac{f_\theta}{f_{\theta'}} - 1\Big|\Big]\mathrm{d}s\,.$$

Assumption 7. *$\mathcal{I}_t^{\theta/\theta'} > 0$ for any $\theta \neq \theta' \in \Theta$.*

Remark 2.6. *Assumption 7 can be interpreted in terms of the set $\mathfrak{D}^{\theta/\theta'} := \{y \in \mathbb{R}, f_\theta(y) \neq f_{\theta'}(y)\}$. We exploit in the proofs the fact that, for any $\theta \neq \theta'$, $\big\{s \in [0,t];\, \bar{\mu}_s^{\theta'}\big[\mathfrak{D}^{\theta/\theta'}\big] > 0\big\}$ has positive Lebesgue measure, which is equivalent to $\mathcal{I}_t^{\theta/\theta'} > 0$.*

Our next result states the consistency of MLE.

Proposition 2.7. *Under Assumptions 1, 2, 4 and 7,*

$$\hat{\theta}_t^N \longrightarrow \theta^* \;\; \textit{in} \;\; \mathbb{P}_N^{\theta^*} - \textit{probability.}$$

This proposition is proved in Section 6.1.

We next show that MLE is asymptotically normal, and under LAN we show its efficiency and local minimax optimality. In order to formulate these results we need to require an additional regularity condition on $\theta \mapsto f_\theta$.

Assumption 8. *For all $x \in \mathbb{R}$, the function $\theta \mapsto f_\theta(x)$ is twice continuously differentiable, and the Hessian matrix, denoted by $\ddot{f}_\theta(x) \in \mathcal{M}^d$, satisfies a uniform bound: $\sup_{\theta\in\Theta}\sup_{x\in\mathbb{R}}\|\ddot{f}_\theta(x)\| < \infty$.*

Theorem 2.8. *Under Assumptions 1 to 8,*

(i) ***Asymptotic normality:*** *For any $\theta^* \in \mathring{\Theta}$, under $\mathbb{P}_N^{\theta^*}$, the following convergence in law holds*

$$\sqrt{N}\big(\hat{\theta}_t^N - \theta^*\big) \xrightarrow{\mathcal{L}} \mathcal{N}\big(0, (I_t^{\theta^*})^{-1}\big)\,.$$

(ii) ***Local asymptotic minimax:*** *The following local asymptotic minimax optimality holds, for any $C > 0$ and any loss functions w that are continuous, bounded and subconvex*

$$\lim_{N\to\infty}\sup_{\sqrt{N}|\theta'-\theta^*|\le C}\mathbb{E}_{\mathbb{P}_N^{\theta'}}\Big[w\Big(N^{1/2}(I_t^{\theta^*})^{1/2}\big(\hat{\theta}_t^N - \theta'\big)\Big)\Big] = \frac{1}{(\sqrt{2\pi})^d}\int_{\mathbb{R}^d} w(x)\,e^{-\frac{1}{2}\|x\|^2}\mathrm{d}x\,.$$

The last Theorem 2.9 is inspired by the paper [15] and their approach based on the results of [29, Section III.1]. In this Theorem we consider a simplified model (1), in the absence of large jumps, i.e. with $\varphi \equiv 0$

$$\mathrm{d}X_t^{i,N,\theta} = b(X_t^{i,N,\theta})\mathrm{d}t + \frac{1}{N}\sum_{j\neq i}^N\int_{\mathbb{R}_+}\int_{\mathbb{R}} u\mathbf{1}_{\{z\le f_\theta(X_{s^-}^{j,N,\theta})\}}\pi^j(\mathrm{d}t,\mathrm{d}z,\mathrm{d}u); \qquad X_0^{i,N,\theta} = X_0^i$$

and its large population limit

$$\mathrm{d}\bar{X}_t^{i,\theta} = b(\bar{X}_t^{i,\theta})\mathrm{d}t + m\,\bar{\mu}_t^{\theta}[f_\theta]\mathrm{d}t; \quad \bar{X}_0^{i,\theta} = X_0^i\,.$$

This approach is based on the following polynomial deviation inequality in the Wasserstein 1 distance between the empirical measure μ^{N,θ^*} of the finite system (1) and the law $\bar{\mu}^{\theta^*}$ of the limit system, see Lemma 6.3, which hold for any $r > 2$, for some $C > 0$ that only depends on r and on the upper-bound on polynomial moments of $\bar{\mu}_s^{\theta^*}$ (up to $2r$)

$$\sup_{s\in[0,t]} \mathbb{E}_{\mathbb{P}_N^{\theta^*}}[\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar{\mu}_s^{\theta^*})^r] \le \frac{C}{N^{r/2}}.$$

For this result to hold, we need the following assumption.

Assumption 9. *(i) For all $r \ge 1$, $\mathbb{E}[|X_0^1|^r] < \infty$.*
(ii) The measure ν satisfies: $\int_{\mathbb{R}} |u|^r \nu(du) < \infty$ for all $r \ge 1$.

The idea behind is to exploit a diagonal coupling with the same Poisson measure for the j-th particle and control together $|X_s^{i,N,\theta^*} - \bar{X}_s^{i,N,\theta^*}|^r$ and $\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar{\mu}_s^{N,\theta^*})^r$ by means of the Grönwall Lemma. More precisely, the Wasserstein norm appears as the natural quantity to bound the predictible part of the interaction term, together with a correction term which we handle with concentration inequalities taken from [20], while the martingale terms are controlled by the Burckholder-Davis-Gundy inequality. This last control fails to produce satisfying bounds if $\varphi \neq 0$, due to the dependency of the jump rate on the state of the particle. We conjecture that such a strong result actually does not hold in this case where $\varphi \neq 0$.

Theorem 2.9 (Convergence and optimality of the MLE without large jumps)**.**

Under Assumptions 1 to 7 and 9, the following asymptotic properties hold:

(i) ***Local asymptotic minimax optimality:*** *For every polynomial loss function w, and any $\delta > 0$ sufficiently small,*

$$\limsup_{N\to\infty} \sup_{|\theta-\theta^*|\le\delta} \mathbb{E}_{\mathbb{P}^\theta}\Big[w\Big(N^{1/2}(I_t^{\theta^*})^{1/2}\big(\hat{\theta}_t^N - \theta\big)\Big)\Big] \to \frac{1}{(\sqrt{2\pi})^d}\int_{\mathbb{R}^d} w(x)\, e^{-\frac{1}{2}\|x\|^2}\mathrm{d}x$$

as $\delta \to 0$.

(ii) ***Global asymptotic minimax optimality:*** *For any non-empty open set $\Theta_0 \subset \mathring{\Theta}$ and every polynomial loss function w,*

$$\mathcal{R}_w^N(\hat{\theta}_t^N; \Theta_0) = \inf_{\tilde{\theta}_N} \mathcal{R}_w^N(\tilde{\theta}_N; \Theta_0)\,(1 + o(1)), \quad \text{as } N \to \infty\,,$$

where the risk is defined as

$$\mathcal{R}_w^N(\tilde{\theta}_N; \Theta_0) = \sup_{\theta\in\Theta_0} \mathbb{E}_{\mathbb{P}^\theta}\Big[w\Big(N^{1/2}(I_t^{\theta})^{1/2}\big(\tilde{\theta}_N - \theta\big)\Big)\Big]\,.$$

Remark 2.10. *The minimax bound in Theorem 2.8 holds for a narrower constraint on θ' and a stronger bound on w as compared to the one in Theorem 2.9.(ii), in line with the conditions in [27] which we draw upon in the present subsection.*

Remark 2.11. *Under the LAN property, we know from [29, Section II.12.1], originally due to Hájek [24], that the r.h.s. of (i) is actually the minimax lower-bound corresponding to the loss function w among any family of estimators.*

3. Numerical illustration of LAN and MLE behavior

3.1. **Simulation setup.** Following Remark 2.1, Equation 1 has been considered with $b(x) = -\alpha x$ corresponding to an exponential leakage at rate α, $\phi(x) = -x$ corresponding to the reset of the neuron state at spiking time, ν the uniform distribution over some set $[\nu_1 - \nu_2, \nu_1 + \nu_2]$ and the following sigmoid function for the spiking rate:

$$f_\theta(x) = \frac{\theta_0}{1 + \exp(-\theta_2 \cdot (x - \theta_1))}; \quad \theta = (\theta_0, \theta_1, \theta_2),$$

where θ_0 corresponds to the maximum intensity, θ_1 to the threshold value of potential, where f_θ has its inflexion point, and θ_2 to the roughness of the transition of f_θ at θ_1.

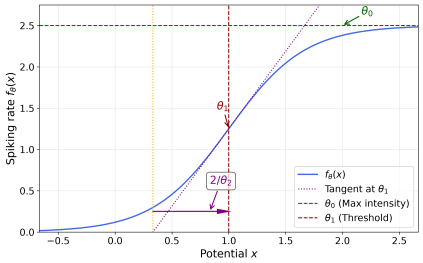


Figure 1. Spiking rate

The parameter values chosen as reference are given in Table 1 for the model parameters and Table 2 for the context of observation. A persisting neuronal activity on the interval $[0, T]$ was observed, as planned, for these parameters. Though one could discern a form of stabilization towards stationnarity, the internal randomness of the system for this neuron number makes each realization unique in terms of the global activity, as detailed in Section 3.2. The choice of parameters was also adapted so that the three parameters θ_0, θ_1 and θ_2 could be inferred with a similar efficiency, which we qualify thanks to the numerical evaluation of the Fisher information matrix $\hat{I}_t^{N,\theta^*}$. This is discussed in next Section 3.4, while we evaluate the prediction derived from the LAN approximation.

$\varphi(x)$	$b(x)$	$\nu(\mathrm{d}u)$	α	ν_1	ν_2	θ_0	θ_1	θ_2
$-x$	$-\alpha x$	$\mathcal{U}(\nu_1-\nu_2, \nu_1+\nu_2)$	1.5	2.4	$0.8 = \nu_1/3$	2.5	1.0	3.0

Table 1. Reference model parameters used in the simulations.

N	T	Δt	X_0^i, $i \in \{1, \dots, N\}$
50	20.0 s	0.002 s	$\mathcal{U}(2, 4)$

Table 2. Reference context of observation.

The dynamics was implemented with a Python implementation customized for the facts that the exponentional leakage is easy to implement and that jumps occur scarcely for each neuron at each small time-step, but reasonably as a whole. The algorithm combines the Lie-Trotter approach in that it alternates between the jumps and the drift, and the Sellke construction [35] for the identification of the neurons that jump at each step. Besides the interest of keeping the random number generation to its bare minimum, the construction moreover produces a natural ordering according to which we can make the neurons jump. Each neuron is treated in order, first with the common random interaction effect, then with the reset of its potential. More details are provided in Appendix A.

3.2. **Visual evidence of persistent stochasticity.** To demonstrate the strength of the proposed approximation, we evaluate it in a context of observation for which stochasticity is visibly not entirely

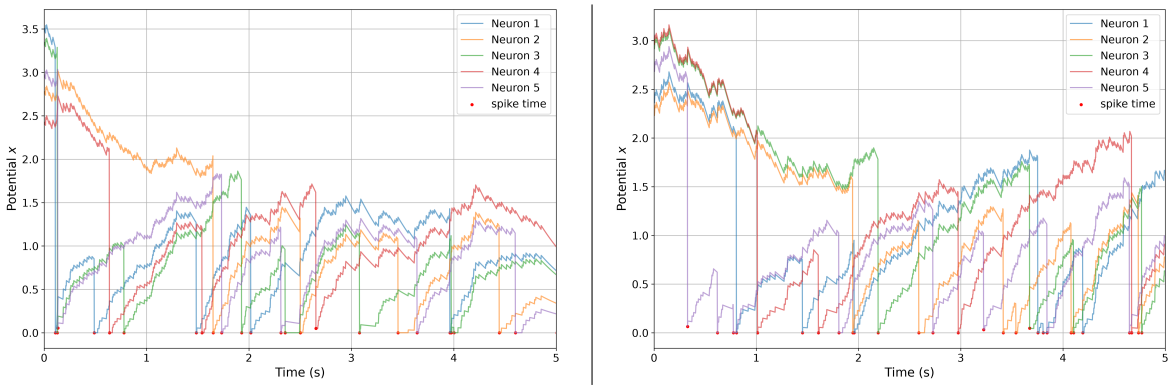


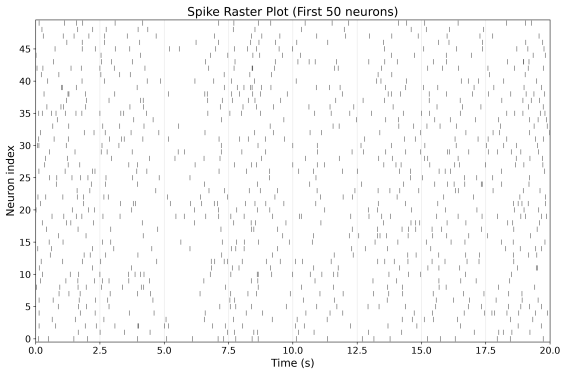
FIGURE 2. Illustration of 5 neuron state trajectories over the time-window $[0, 5]$, for two replicates.

averaged out, though the system approaches a stationary regime. More regular scenarios are also displayed in Appendix Section A.2.1, for $N = 100$ and $N = 500$ neurons.

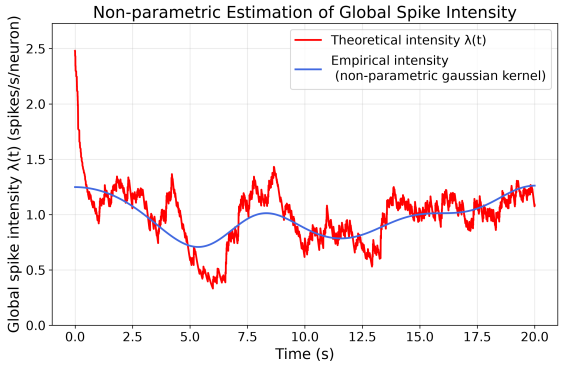
3.2.1. *Individual neuron trajectories.* The randomness of the system can be observed first by looking at neuron trajectories, which we show in Figure 2 for 5 neuron trajectories with two replicates of the same system (one per subpanel). Even in the limiting regime, randomness persists at the neuronal level due to the spike events, that trigger neuron potential reset. These events are easily identified under the proposed scenario and occur several times for each neuron. Although the limiting drift presumably accounts for most of the dynamics between these events, shared fluctuations introduce significant deviations from the deterministic flow across all neurons.

3.2.2. *Spike raster plot and global intensity.* In Figure 3, we additionally present an analysis of the global spiking intensity. The spiking times for each neuron is displayed in subpanel (A). We can observe globally that the spike events do not occur uniformly and that some time-periods appear to be reduced (between 5s and 7s) or on the contrary enriched (between 7.5s and 8s) in spike events. We exploited a nonparametric gaussian kernel approach for estimating the spiking intensity of the whole system. The outcome is presented in blue in subpanel (B), and compared with the expected spike intensity (in red) that we can derive by computing the spiking rate f_θ over the empirical distribution of neuron potentials. The blue curve appears to confirm the visual observation of fluctuations, though it is less confirmed than with the red curve. The red curve shows that the depletion is expected from the empirical distribution of neuron states. While the activity might have been larger than this expectation, the kernel bandwidth, adjusted for spike event randomness, may be the main reason for the smaller decrease of the blue curve.

3.3. **Statistical validation of LAN quantities.** To evaluate the LAN property as stated in Theorem 2.4, we focus on the empirical behavior of the random sequence Δ_t^{N,θ^*} and the empirical Fisher information matrix I_t^{N,θ^*}, given respectively by Equations 13 and 15, across the 100 replicates. The LAN property predicts that Δ_t^{N,θ^*} should converge in law to a Gaussian distribution $\mathcal{N}(0, I_t^{\theta^*})$, with the empirical Fisher information I_t^{N,θ^*} converging to its theoretical counterpart $I_t^{\theta^*}$, defined in Equation 6.



(A) Raster plot of the spike trains for all 50 neurons. Time variations of the spike intensity can be identified.



(B) Estimation of the global spiking intensity, in blue with a nonparametric gaussian kernel intensity derived from the spike times, in red with the computation of the spiking rate over the neuronal states (for comparison).

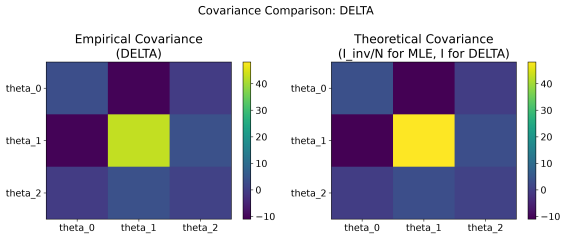
FIGURE 3. Study of the spiking times.

Because it appears of practical interest to characterize the variance of the MLE in terms of an estimator from the observed data, we will make a discrete computation of $\widehat{I}_t^N = I_t^{N,\hat{\theta}_t^N}$ as the estimated Fisher information. Since we want to select only one value $\widehat{I}_t^N$ among the set of replicates, the chosen one corresponds to a typical replicate. The typical replicate is set as the one for which the relative error in Mahalanobis distance is exactly at the median value, see Figure 5-(B) explanations.

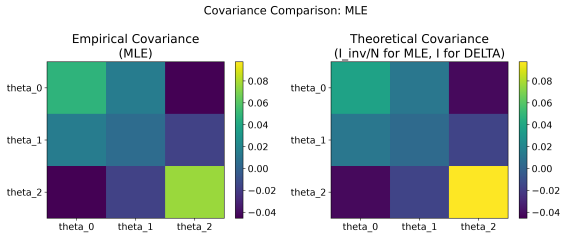
The difference in Frobenius norm as compared to I_t^{N,θ^*} has been observed at around 5%. This in strong constrast with the relative difference between the discrete computation of $\Delta_t^{N,\hat{\theta}_t^N}$ and of Δ_t^{N,θ^*}, the former being very close to 0 (which we can explain by relating $\Delta_t^{N,\hat{\theta}_t^N}$ to the derivative of the log-likelihood at $\hat{\theta}_t^N$). Only the later is considered in the following.

The alignment between the empirical and theoretical covariance structures is first illustrated in Figure 4-(A), which compares the empirical covariance matrix of Δ_t^{N,θ^*} (computed over the 100 replicates) with the theoretical Fisher information matrix $\widehat{I}_t^N$. The visual similarity between the two matrices is confirmed by a relative Frobenius norm error of 11%. On the right panel, a similar analysis is conducted for the empirical covariance matrix of the MLE as compared to $N^{-1}(\widehat{I}_t^N)^{-1}$. The relative Frobenius norm error is at 20%, which is good yet not entirely satisfying for computing quasi-confidence intervals.

We conducted three multivariate goodness-of-fit tests for Δ_t^{N,θ^*}. The Henze-Zirkler test for multivariate normality [26] does not contradict the null hypothesis of normality (p-value of 0.45). Then, Hotelling's T^2 [28] test does not reject either the null hypothesis of a null mean vector (p-value of 0.71). Finally, Mauchly's test [32] for sphericity was exploited after correcting the covariance matrix by considering $(\widehat{I}_t^N)^{-1/2}\Delta_t^{N,\theta^*}$. The covariance structure is also far from being rejected (p-value of 0.20). So all three tests pass, providing strong empirical evidence that the distribution of Δ_t^{N,θ^*} aligns with the theoretical predictions of the LAN property for $N = 50$.



(A) Empirical covariance matrix of Δ_t^{N,θ^*} (over 100 replicates) as compared to $\widehat{I}_t^N$. Relative error in Frobenius norm: 0.11.



(B) Empirical covariance matrix of the MLE as compared to $N^{-1}(\widehat{I}_t^N)^{-1}$. Relative error in Frobenius norm: 0.20.

FIGURE 4. Comparison between the empirical covariance and the prediction with the estimated Fisher information matrix.

The same tests were then applied to the MLE. Normality is confirmed (p-value of 0.52 at the Henze-Zirkler test), as well as the mean value at θ^* (p-value of 0.52 at the Hotelling's T^2 test), and the projected covariance matrix (p-value of 0.77 at Mauchly's test applied to $N^{1/2}(\widehat{I}_t^N)^{1/2}\,(\hat{\theta}_t^N - \theta^*)$).

3.4. **MLE performance and alignment with the LAN predictor.** In Figure 5-(A), we compare the histograms derived from the MLE $\hat{\theta}_t^N$ and the LAN predictor $\check{\theta}_t^N = \theta^* + (\widehat{I}_t^{N,\theta^*})^{-1}\Delta_t^{N,\theta^*}/\sqrt{N}$. It demonstrates that they have very similar distributions, close to normality and with very similar intervals of fluctuation (`Mean`$\pm$`2Std`).

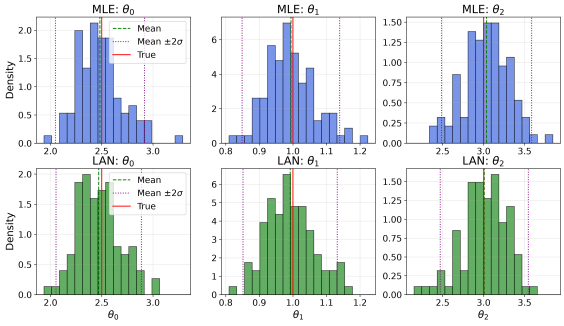
To quantify more precisely how closely the MLE is well-described by the LAN predictor in finite-N regimes, we analyze the relative errors between the two quantities as compared to θ^* across a set of 100 simulations. The relative error is defined as $\|\hat{\theta}_t^N - \check{\theta}_t^N\|/\|\hat{\theta}_t^N - \theta_*\|$. A small relative error is indicative that the higher-order terms in entail indeed second order corrections for the MLE, and that the numerical estimation has been efficient.

Figure 5-(B) displays the cumulative distribution function (c.d.f.) of the relative error across the 100 simulations, in blue line for the Euclidian distance. It is also natural to check the quality of estimation when we take into account the expected covariance matrix, as it was done e.g. in Corollary 2.5. This corresponds to computing the relative error with respect to a relevant Mahalanobis distance $\|.\|_M$, where $\|\theta\|_M^2 = N^1\theta^\top\,\widehat{I}_t^{N,\theta^*}\,\theta^1$. As we see in Figure 5-(B), the LAN predictor appears to fit closely the MLE for both distances, with very similar distributions. 50% of the replicates have a relative error below 10%, and close to 90% of them below 20%, for both distances.

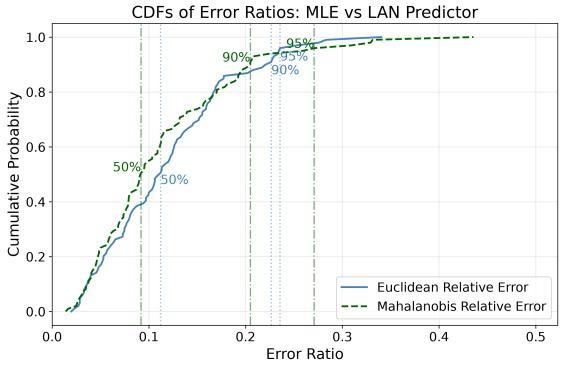
For some first sets of parameters, arbitrarily chosen, the relative error in Euclidian distance was much larger. We suspected it could be partly due to the fact that one of the parameters (often θ_2) is much worse estimated than the others, leading to larger deviations between $\hat{\theta}_t^N$ and $\check{\theta}_t^N$ in this direction as compared to θ_*. This was our original motivation for computing the relative error in terms of the Mahalanobis distance $\|.\|_M$.

3.5. **Conclusion of the simulations.** This numerical evaluation complements the theoretical results of Section 2 and demonstrates that the LAN approximation remains accurate even for moderate population sizes where stochastic fluctuations are still visible.

[1]In computing the relative error, the scaling factor N plays no role and can be forgotten



(A) Histograms of the MLE (top row) and LAN predictor (bottom row) for the three parameters θ_0, θ_1, and θ_2. The red lines indicate the true parameter values $\theta^* = [2.5, 1.0, 3.0]$, the dashed green line the empirical mean and the two orange dotted lines the interval of fluctuations (Mean $\pm 2\sigma$) .



(B) C.d.f. of the relative error $\|\hat{\theta}_t^N - \check{\theta}_t^N\|/\|\hat{\theta}_t^N - \theta_*\|$ for the Euclidian distance (blue line) and the Mahalanobis distance (green dashed line). Vertical lines display the quantiles of both distributions corresponding to 50%, 90% and 95%.

FIGURE 5. Comparison between the estimated MLE $\hat{\theta}_t^N$ and the LAN predictor $\check{\theta}_t^N = \theta^* + (\widehat{I}_t^{N,\theta^*})^{-1}\Delta_t^{N,\theta^*}/\sqrt{N}$, as compared to θ^*.

4. Auxiliary results

This section gathers results used to establish the LAN property and to analyze the Maximum Likelihood Estimator.

4.1. **Continuity.**

Proposition 4.1. *Under Assumptions 2-3, for any continuous and bounded function $g : \mathbb{R} \to \mathbb{R}$, the following convergence holds in $P_N^{\theta^*}$ - probability*

$$\forall t > 0, \quad \int_0^t \mu_s^{N,\theta^*}(g)ds \to \int_0^t \bar{\mu}_s^{\theta^*}(g)ds.$$

Proof. We assume that the family of i.i.d. Poisson random measures $(\pi_i)_{i\geq 1}$, is defined on some filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_t, \mathbb{P})$ and that for each $\theta \in \Theta$ the strong solution of (3) is constructed. The law on $\mathcal{D}(\mathbb{R}_+, \mathbb{R})$ equipped with the natural filtration of this solution is denoted by $\bar{\mu}^\theta$. We know that μ^{N,θ^*} converges in law in $\mathcal{P}(\mathcal{D}(\mathbb{R}_+, \mathbb{R}))$ to $\bar{\mu}^{\theta^*}$ under Assumptions 2-3, see [5].

Let $g : \mathbb{R} \to \mathbb{R}$ be continuous and bounded, $t > 0$ fixed and $F_g^t : \mathcal{P}(\mathcal{D}(\mathbb{R}_+, \mathbb{R})) \to \mathbb{R}$ given by

$$\mathcal{P}(\mathcal{D}(\mathbb{R}_+, \mathbb{R})) \ni m \mapsto \int_0^t m_s(g)ds.$$

Using Billingsley, for fixed $s > 0$, the projection application $\mathcal{P}(\mathcal{D}(\mathbb{R}_+, \mathbb{R})) \ni m \mapsto m_s \in \mathcal{P}(\mathbb{R})$ is continuous in all point $\tilde{m} \in \mathcal{P}(\mathcal{D}(\mathbb{R}_+, \mathbb{R}))$, such that

$$\tilde{m}(\{\gamma \in \mathcal{D},\ \gamma_s \neq \gamma_{s-}\}) = 0.$$

It is easy to see that the law of the limit McKean-Vlasov equation satisfy this property, since

$$\bar{\mu}^{\theta^*}(\{\gamma \in \mathcal{D},\ \gamma_s \neq \gamma_{s-}\}) \leq \mathbb{P}(\pi(s \times [0, B_0]) > 0) = 0.$$

Using the continuous mapping theorem, if m^n converges in law to m, then m_s^n converges in law to m_s, which means that $\forall g \in C_b(\mathbb{R})$, $m_s^n(g) \to m_s(g)$. We conclude by dominated convergence theorem that $\int_0^t m_s^n(g)ds \to \int_0^t m_s(g)ds$. As a result F_g^t is continuous in $\bar{\mu}$ and the proposition follows. □

4.2. **Handleable likelihood.** Let us give a more handleable expression of the likelihood. Denote $\bar{\pi}^j(\mathrm{d}t, \mathrm{d}z) = \int_{\mathbb{R}} \pi^j(\mathrm{d}t, \mathrm{d}z, \mathrm{d}u)$ and as usually denote $\widetilde{\pi}^j$ the compensated measure:

$$\widetilde{\pi}^j(\mathrm{d}t, \mathrm{d}z, \mathrm{d}u) = \pi^j(\mathrm{d}t, \mathrm{d}z, \mathrm{d}u) - \mathrm{d}s\, \mathrm{d}z\, \nu(\mathrm{d}u), \quad \widetilde{\bar{\pi}}^j(\mathrm{d}t, \mathrm{d}z) = \int_{\mathbb{R}} \widetilde{\pi}^j(\mathrm{d}t, \mathrm{d}z, \mathrm{d}u)\,.$$

Define

$$J^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x) = \sum_{j=1}^{N} \sum_{n\geq 1} \delta_{\left(T_n^j, X^{j,N,\theta^*}_{T_n^j-}\right)}(\mathrm{d}s, \mathrm{d}x), \tag{8}$$

in which all jump times and corresponding neuron potentials are recorded, and the related compensated measure:

$$\widetilde{J}^{N,\theta^*} = J^{N,\theta^*} - \Upsilon^{N,\theta^*}, \quad \text{with } \Upsilon^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x) = N\, f_{\theta^*}(x)\, \mu_s^{N,\theta^*}(\mathrm{d}x)\, \mathrm{d}s\,, \tag{9}$$

and

$$\mu_s^{N,\theta^*}(\mathrm{d}x) = \frac{1}{N} \sum_{j=1}^{N} \delta_{X_s^{j,N,\theta^*}}(\mathrm{d}x),$$

where μ_s^{N,θ^*} denotes the empirical measure of the system $(X^{i,N,\theta^*})_{i\leq N}$ at time t.

Therefore

$$\log L_t^{\theta/\theta^*} = \int_{[0,t]\times\mathbb{R}} \log\left(\frac{f_\theta}{f_{\theta^*}}\right)(x)\; J^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x) - \int_{[0,t]\times\mathbb{R}} \left(\frac{f_\theta}{f_{\theta^*}}(x) - 1\right) \Upsilon^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x)\,, \tag{10}$$

which can also be rewritten as

$$\log L_t^{\theta/\theta^*} = M_t^{N,\theta/\theta^*} + A_t^{N,\theta/\theta^*}\,, \tag{11}$$

where

$$M_t^{N,\theta/\theta^*} := \iint_{[0,t]\times\mathbb{R}} \left(\log \frac{f_\theta}{f_{\theta^*}}\right)(x)\; \widetilde{J}^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x)$$

and

$$\begin{aligned} A_t^{N,\theta/\theta^*} &:= \int_0^t \int_{\mathbb{R}} \left(\log \frac{f_\theta}{f_{\theta^*}} - \frac{f_\theta}{f_{\theta^*}} + 1\right)(x)\; \Upsilon^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x) \\ &= N \int_0^t \mu_s^{N,\theta^*} \Big[\Big(\log \frac{f_\theta}{f_{\theta^*}} - \frac{f_\theta}{f_{\theta^*}} + 1\Big) \cdot f_{\theta^*}\Big]\, \mathrm{d}s. \end{aligned}$$

Now, put $\theta = \theta^* + h/\sqrt{N}$ and define

$$M_t^{N,\theta^*/\theta^*} := h\, \Delta_t^{N,\theta^*} + \widetilde{M}_t^{N,\theta/\theta^*}\,, \tag{12}$$

with

$$\begin{aligned} \Delta_t^{N,\theta^*} &:= \frac{1}{\sqrt{N}} \iint_{[0,t]\times\mathbb{R}} \frac{\dot{f}_{\theta^*}}{f_{\theta^*}}(x)\; \widetilde{J}^{N,\theta^*}(\mathrm{d}s, \mathrm{d}x) \\ &= \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \iint_{[0,t]\times\mathbb{R}^+} \frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\big(X_{s-}^{j,N,\theta^*}\big) \mathbf{1}_{\left\{z\leq f_{\theta^*}\left(X_{s-}^{j,N,\theta^*}\right)\right\}} \widetilde{\bar{\pi}}^j(\mathrm{d}s, \mathrm{d}z) \end{aligned} \tag{13}$$

and

$$\widetilde{M}_t^{N,\theta/\theta^*} := \iint_{[0,t]\times\mathbb{R}} \left(\log \frac{f_\theta}{f_{\theta^*}} - (\theta-\theta^*)^\intercal \cdot \frac{\dot{f}_{\theta^*}}{f_{\theta^*}} \right)(x)\ \tilde{J}^{N,\theta^*}(\mathrm{d}s,\mathrm{d}x)$$
$$= \sum_{j=1}^{N} \int_{[0,t]\times\mathbb{R}_+} \left(\log \frac{f_\theta}{f_{\theta^*}} - (\theta-\theta^*)^\intercal \cdot \frac{\dot{f}_{\theta^*}}{f_{\theta^*}} \right)\left(X_{s-}^{j,N,\theta^*}\right)\mathbf{1}_{\left\{z\le f_{\theta^*}\left(X_{s-}^{j,N,\theta^*}\right)\right\}}\widetilde{\bar{\pi}}^j(\mathrm{d}s,\mathrm{d}z)\,.$$

Letting $g : y \in (0,\infty) \mapsto \log(1+y) - y$, we observe that $g(y) \sim -\frac{1}{2}y^2$ for y in the vicinity of 0. This explains the interest in the following decomposition of $A_t^{N,\theta/\theta^*}$:

$$A_t^{N,\theta/\theta^*} = \frac{-1}{2}\, h^\intercal I_t^{N,\theta^*} h + \widetilde{A}_t^{N,\theta/\theta^*}, \tag{14}$$

where

$$I_t^{N,\theta^*} = \frac{1}{N}\iint_{[0,t]\times\mathbb{R}} \left(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\right)^{\otimes 2}(x)\ \Upsilon^{N,\theta^*}(\mathrm{d}s,\mathrm{d}x) = \int_0^t \mu_s^{N,\theta^*}\left[\frac{(\dot{f}_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\right]\mathrm{d}s \tag{15}$$

and

$$\widetilde{A}_t^{N,\theta/\theta^*} = \iint_{[0,t]\times\mathbb{R}} \left(g\circ\left(\frac{f_\theta}{f_{\theta^*}}-1\right) + \frac{1}{2}\left[(\theta-\theta^*)^\intercal\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\right]^2\right)(x)\ \Upsilon^{N,\theta^*}(\mathrm{d}s,\mathrm{d}x)$$
$$= N\int_0^t \mu_s^{N,\theta^*}\left[\left(g\circ\left(\frac{f_\theta}{f_{\theta^*}}-1\right) + \frac{1}{2}\left[(\theta-\theta^*)^\intercal\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\right]^2\right)\cdot f_{\theta^*}\right]\mathrm{d}s\,.$$

5. Proof of the LAN property

Proposition 5.1. *Under Assumptions 1 to 5, for all $t > 0$, the following convergences hold, as $N \to \infty$,*

(i) $I_t^{N,\theta^*} \longrightarrow I_t^{\theta^*}$ *in $\mathbb{P}_N^{\theta^*}$-probability;*
(ii) $\Delta_t^{N,\theta^*} \longrightarrow \mathcal{N}(0, I_t^{\theta^*})$ *in distribution under $\mathbb{P}_N^{\theta^*}$;*
(iii) $\widetilde{A}_t^{N,\theta/\theta^*} \longrightarrow 0$ *in $\mathbb{P}_N^{\theta^*}$-probability;*
(iv) $\widetilde{M}_t^{N,\theta/\theta^*} \longrightarrow 0$ *in $\mathbb{P}_N^{\theta^*}$-probability.*

Proof. (i) Using the representation

$$I_t^{N,\theta^*} = \int_0^t \mu_s^{N,\theta^*}\left[\frac{(\dot{f}_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\right]\mathrm{d}s$$

with $(\dot{f}_{\theta^*})^{\otimes 2}/f_{\theta^*}$ continuous and bounded by $B_1^2 B_0$ according to Assumption 4, item (i) follows from Proposition 4.1.

(ii) Observe that this result can be derived from a following more general statement:

$$(\Delta_s^{N,\theta^*})_{s\ge 0} \to \left(\int_0^s \left(\bar{\mu}_r^{\theta^*}\left[\frac{(\dot{f}_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\right]\right)^{1/2}\mathrm{d}B_r\right)_{s\ge 0},$$

in distribution in $\mathcal{D}(\mathbb{R}_+,\mathbb{R})$, as $N\to\infty$. Here $(B_s)_{s\ge 0}$ denotes a standard d-dimensional Brownian motion. To prove this last assertion we use martingale convergence theorem VIII.3.22 of [30], therefore

we need to satisfy the following convergences in $\mathbb{P}_N^{\theta^*}$ probability

$$\int_{\mathbb{R}_+}\int_{\mathbb{R}} \frac{(\dot f_{\theta^*})^{\otimes 2}}{f_{\theta^*}}(x)\,\mathbf{1}_{\left\{\left\|\frac{\dot f_{\theta^*}}{f_{\theta^*}}(x)\right\|>\varepsilon\sqrt{N}\right\}}\mu_s^{N,\theta^*}(\mathrm{d}x)\,\mathrm{d}s\to 0,\quad \forall\varepsilon>0\,, \tag{16}$$

and Condition $[\gamma' 5-D]$ in [30, p.432]

$$<\Delta^{N,\theta^*}>_t\to I_t^{\theta^*}, \tag{17}$$

where the quadratic variation is identified as follows:

$$<\Delta^{N,\theta^*}>_t=\frac{1}{N}\iint_{[0,t]\times\mathbb{R}}\Big(\frac{\dot f_{\theta^*}}{f_{\theta^*}}\Big)^{\otimes 2}(x)\,\Upsilon^{N,\theta^*}(\mathrm{d}s,\mathrm{d}x)=\int_0^t\mu_s^{N,\theta^*}\Big[\frac{(\dot f_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\Big]\,\mathrm{d}s=I_t^{N,\theta^*}\,.$$

Therefore (17) is already proved in point (i). With $\|\dot f_{\theta^*}/f_{\theta^*}\|$ bounded by B_1, (16) is straightforward, since for N sufficiently large the indicator function of the event $\left\|\frac{\dot f_{\theta^*}}{f_{\theta^*}}(x)\right\|>\varepsilon\sqrt{N}$ is zero. It concludes the proof of the point (ii).

(iii) We remark that $g(y)+(1/2)\,y^2=O(y^3)$ as y tends to 0. For $\frac{\dot f_{\theta^*}}{f_{\theta^*}}$ is bounded, Assumption 5 entails that $\Big(\frac{f_\theta}{f_{\theta^*}}-1\Big)$ is sufficiently small for N large enough so that the following holds for some $C_1,C_3,C_g>0$:

$$\Big\|\frac{f_\theta}{f_{\theta^*}}-1\Big\|_\infty\le C_1\,\|\theta-\theta^*\|,\qquad \Big\|\Big(\frac{f_\theta}{f_{\theta^*}}-1\Big)^2-\Big((\theta-\theta^*)^\intercal\frac{\dot f_{\theta^*}}{f_{\theta^*}}\Big)^2\Big\|_\infty\le C_3\,\|\theta-\theta^*\|^3\,,$$
$$\Big\|g\circ\Big(\frac{f_\theta}{f_{\theta^*}}-1\Big)+\frac{1}{2}\Big((\theta-\theta^*)^\intercal\frac{\dot f_{\theta^*}}{f_{\theta^*}}\Big)^2\Big\|_\infty\le C_g\,\|\theta-\theta^*\|^3=O\Big(\frac{1}{N\sqrt{N}}\Big)\,.$$

Since f_{θ^*} is bounded and μ_s^{N,θ^*} has total mass 1, it concludes the proof of (iii).

(iv) We prove it by justifying the convergence to 0 of the quadratic variation:

$$\begin{aligned}<\widetilde{M}^{N,\theta/\theta^*}>_t&=\iint_{[0,t]\times\mathbb{R}}\Big(\log\big[\frac{f_\theta}{f_{\theta^*}}(x)\big]-(\theta-\theta^*)^\intercal\cdot\frac{\dot f_{\theta^*}}{f_{\theta^*}}(x)\Big)^2\,\Upsilon^{N,\theta^*}(\mathrm{d}s,\mathrm{d}x)\\&=N\int_0^t\mu_s^{N,\theta^*}\Big[\Big(\log\big[\frac{f_\theta}{f_{\theta^*}}\big]-(\theta-\theta^*)^\intercal\cdot\frac{\dot f_{\theta^*}}{f_{\theta^*}}\Big)^2\cdot f_{\theta^*}\Big]\,\mathrm{d}s\,.\end{aligned}$$

In the same spirit as for point (iii), there exists $C_2>0$ such that:

$$\Big\|\log\big[\frac{f_\theta}{f_{\theta^*}}\big]-(\theta-\theta^*)^\intercal\cdot\frac{\dot f_{\theta^*}}{f_{\theta^*}}\Big\|_\infty\le C_2\,\|\theta-\theta^*\|^2=O(1/N)\,.$$

Therefore $<\widetilde{M}^{N,\theta/\theta^*}>_t=O(1/N)$ which entails that the martingale $\widetilde{M}^{N,\theta/\theta^*}$ converges to 0 under $\mathbb{P}_N^{\theta^*}$, as $N\to\infty$. □

Proof of Theorem 2.4. Using the decomposition (11) together with decompositions (12), (14), we can write

$$\log L_t^{\theta/\theta^*}=h\Delta_s^{N,\theta^*}-\frac{1}{2}h^\intercal I_t^{N,\theta^*}h+r_N(\theta,\theta^*)\,,$$

where

$$r_N(\theta,\theta^*)=\widetilde{M}_t^{N,\theta/\theta^*}+\widetilde{A}_t^{N,\theta/\theta^*}.$$

The proof is complete with the results of Proposition 5.1 telling us that both $\widetilde{M}_t^{N,\theta/\theta^*}$ and $\widetilde{A}_t^{N,\theta/\theta^*}$ converge to zero in $\mathbb{P}_N^{\theta^*}$-probability, as $N\to\infty$.

□

6. Maximum Likelihood estimator

6.1. **Consistency.** In this section we show the consistency of the MLE.

6.1.1. *Proof of the consistency of the MLE.*

of Proposition 2.7. We first show that

$$\frac{1}{N}\log L_t^{\theta/\theta^*}\big(X_{[0,t]}^{\theta^*,N}\big) \to A_t^{\theta^*}(\theta)$$

in $\mathbb{P}_N^{\theta^*}-$ probability, as $N\to\infty$, where $A_t^{\theta^*}:\Theta\to\mathbb{R}$ is such that $A_t^{\theta^*}(\theta)<0=A_t^{\theta^*}(\theta^*)$ for any $\theta\in\Theta$ with $\theta\neq\theta^*$.

Recall (see LAN subsection) that the log-likelihood ratio process is given by

$$\log L_t^{\theta/\theta^*} = A_t^{N,\theta/\theta^*} + M_t^{N,\theta/\theta^*},$$

where the processes $A_t^{N,\theta/\theta^*}$ and $M_t^{N,\theta/\theta^*}$ are defined in (11). On the one hand, using Proposition 4.1 together with Assumption 4 (i), we get that

$$\frac{1}{N^2}<M^{N,\theta/\theta^*}>_t = \frac{1}{N}\int_0^t \mu_s^{N,\theta^*}\Big[\Big(\log\frac{f_\theta}{f_{\theta^*}}\Big)^2\cdot f_{\theta^*}\Big]\,\mathrm{d}s\to 0\,, \tag{18}$$

which entails that

$$\frac{1}{N}M_t^{N,\theta/\theta^*}\longrightarrow 0 \text{ in } \mathbb{P}_N^{\theta^*}-\text{probability}.$$

For the consistency, we will need though a stronger result to deal with the uniformity in θ for this convergence.

On the other hand, we exploit again Proposition 4.1 since the function $\big(\log\frac{f_\theta}{f_{\theta^*}}-\frac{f_\theta}{f_{\theta^*}}+1\big)\cdot f_{\theta^*}$ is bounded continuous on $\mathbb{R}$, leading to:

$$\begin{aligned}\frac{1}{N}A_t^{N,\theta/\theta^*} &= \int_0^t \mu_s^{N,\theta^*}\Big[\Big(\log\frac{f_\theta}{f_{\theta^*}}-\frac{f_\theta}{f_{\theta^*}}+1\Big)\cdot f_{\theta^*}\Big]\,\mathrm{d}s\\ &\to \int_0^t \bar{\mu}_s^{\theta^*}\Big[\Big(\log\frac{f_\theta}{f_{\theta^*}}-\frac{f_\theta}{f_{\theta^*}}+1\Big)\cdot f_{\theta^*}\Big]\,\mathrm{d}s := A_t^{\theta/\theta^*},\end{aligned} \tag{19}$$

in $\mathbb{P}_N^{\theta^*}-$ probability, as $N\to\infty$. Notice that $\log(r)-r+1\leq 0, \forall r\in\mathbb{R}$, with the equality holding only at $r=1$. Therefore,

$$A_t^{\theta/\theta^*}\leq 0 = A_t^{\theta^*/\theta^*}.$$

In addition, Assumption 7 entails that $A_t^{\theta/\theta'}<0$ for any $\theta\neq\theta'\in\Theta$, see also Remark 2.6. θ^* is thus the only maximum of $\theta\mapsto A_t^{\theta/\theta^*}$.

Secondly, we show that : $\theta\mapsto\log L_t^{\theta/\theta^*}$ is unlikely to vary very abruptly. First note that $J^{N,\theta^*}([0,t]\times\mathbb{R})$ (recall the definitions 8 and (4)) is upper-bounded by

$$J^{N,\theta^*}([0,t]\times\mathbb{R})\leq \bar{J}_t^N := \sum_{j=1}^N \bar{\pi}^j\big([0,t]\times[0,B_0]\big)\,.$$

The law of $\bar\pi^j\big([0,t]\times[0,B_0]\big)$ is Poisson with mean $t\,B_0$, for any j, and these variables are independent. By the strong law of large number,

$$\lim_{N\to\infty} \bar J_t^N/N = t\,B_0, \quad \mathbb{P}-a.s.,$$

where $\mathbb{P}$ is the probability on the original space where (π^j), $j\in\mathbb{N}^*$ are defined. In particular,

$$\lim_{N\to\infty} \mathbb{P}(\mathcal{J}^N) = 1,$$

where

$$\mathcal{J}^N = \{\bar J_t^N/N \le 2t\,B_0\}\,.$$

For any $\theta,\theta'\in\Theta$, using the definition (10), on the event $\mathcal{J}^N$:

$$\frac{1}{N}|\log L_t^{\theta/\theta^*} - \log L_t^{\theta'/\theta^*}| \le \frac{J^{N,\theta^*}([0,t]\times\mathbb{R})}{N}\cdot\Big\|\log\frac{f_\theta}{f_{\theta'}}\Big\|_\infty + t\,\|f_\theta - f_{\theta'}\|_\infty \le C_L\,|\theta-\theta'|\,, \tag{20}$$

with $C_L := t\,(2B_0+1)\,(B_1\vee B_L) > 0$.

Let $\delta,\varepsilon>0$, $B(\theta^*,\varepsilon)$ be the ball in Θ of radius ε and center θ^*, and

$$\eta := \min\big\{|A_t^{\theta/\theta^*}|;\ \theta\in\Theta\setminus B(\theta^*,\varepsilon)\big\}.$$

Because of the previous observation and the continuity of $\theta\mapsto A_t^{\theta/\theta^*}$, η is positive. We can consider N sufficiently large ($N\ge N_0$) to ensure $\mathbb{P}([\mathcal{J}^N]^c)\le\delta/2$, and so restrict our analysis to the event $\mathcal{J}^N$. Let $r=\eta/(4C_L)>0$. Because Θ is compact, there exists a finite sequence $(\theta_\ell)_{\ell\in[\![1,L]\!]}$ of elements of $\Theta\setminus B(\theta^*,\varepsilon)$ such that the balls $\big(B(\theta_\ell,r)\big)_\ell$ cover $\Theta\setminus B(\theta^*,\varepsilon)$. For each $\ell\in[\![1,L]\!]$, (18) and (19), entail that

$$\frac{1}{N}\log L_t^{\theta_\ell/\theta^*}\big(X^{N,\theta^*}_{[0,t]}\big) \longrightarrow A_t^{\theta_\ell/\theta^*} \ \text{ in } \ \mathbb{P}_N^{\theta^*}-\text{probability, as } N\to\infty.$$

We can therefore ensure that $\mathbb{P}_N^{\theta^*}\big(\mathcal{A}^N\big)\ge 1-\frac{\delta}{2}$ holds for any N sufficiently large ($N\ge N_1$), where

$$\mathcal{A}^N := \Big\{\forall \ell\in[\![1,L]\!],\ \Big|\frac{1}{N}\log L_t^{\theta_\ell/\theta^*}\big(X^{N,\theta^*}_{[0,t]}\big) - A_t^{\theta_\ell/\theta^*}\Big| \le \frac{\eta}{4}\Big\}\,.$$

Remark also that $A_t^{\theta_\ell/\theta^*}<-\eta$, for any $\ell\in[\![1,L]\!]$ by definition of η and the sign of A_t^{θ/θ^*}.

Let $\theta\in\Theta\setminus B(\theta^*,\varepsilon)$. There exists $\ell\in[\![1,L]\!]$ such that $\|\theta-\theta_\ell\|\le r$. On the event $\mathcal{J}^N\cap\mathcal{A}^N$, we have, by the Lipschitz property (20), that

$$\frac{1}{N}\Big|\log L_t^{\theta/\theta^*} - \log L_t^{\theta_\ell/\theta^*}\Big| \le C_L\,\|\theta-\theta_\ell\| \le C_L\,r = \frac{\eta}{4}\,.$$

Therefore, for any $\theta\in\Theta\setminus B(\theta^*,\varepsilon)$

$$\frac{1}{N}\log L_t^{\theta/\theta^*} \le A_t^{\theta_\ell/\theta^*} + \frac{\eta}{4} + \frac{\eta}{4} < -\frac{\eta}{2}\,.$$

On the other hand, $\log L_t^{\theta^*/\theta^*}=0$ by definition. Since $\hat\theta_t^N$ is the maximizer of L_t^{θ/θ^*}, we thus have:

$$\frac{1}{N}\log L_t^{\hat\theta_t^N/\theta^*} \ge 0 > \sup\Big\{\frac{1}{N}\log L_t^{\theta/\theta^*};\ \theta\in\Theta\setminus B(\theta^*,\varepsilon)\Big\}\,,$$

on the event $\mathcal{J}^N\cap\mathcal{A}^N$ for N sufficiently large ($N\ge N_0\vee N_1$). It then follows that:

$$\mathbb{P}_N^{\theta^*}\big(|\hat\theta_t^N-\theta^*|\ge\varepsilon\big) \le \mathbb{P}\big([\mathcal{J}^N]^c\big) + \mathbb{P}\big([\mathcal{A}^N]^c\big) \le \delta\,.$$

As ε,δ are freely chosen, this concludes the proof of the consistency. □

6.2. **Limit distribution and local asymptotic minimax optimality of MLE.** In this section we show the asymptotic normality and the local asymptotic minimax optimality of MLE. The proof follows the arguments of [27, (7.12)]. In particular we show that MLE satisfies the "coupling property" of [27, (7.9)].

6.2.1. *Proof of Theorem 2.8.*

Proof. Denote by $\ell^N[\theta] = \log L_t^{\theta/\theta^*}$, then using Taylor formula and the fact that $\hat{\theta}^N$ is the argmax of ℓ^N:

$$-\dot{\ell}^N[\theta^*] = \Big[\int_0^1 \ddot{\ell}^N[\theta^* + \xi\cdot(\hat{\theta}_t^N - \theta^*)]\,\mathrm{d}\xi\Big]\cdot(\hat{\theta}_t^N - \theta^*), \tag{21}$$

where $\dot{\ell}^N$ and $\ddot{\ell}^N$ denote respectively the gradient vector and the Hessian matrix of ℓ^N. We identify that $N^{-1/2}\dot{\ell}^N[\theta^*]$ is equal to the term Δ_t^{N,θ^*} from the LAN decomposition, given by (12), and recall that this martingale satisfies $< \Delta^{N,\theta^*} >_t = I_t^{N,\theta^*}$. The convergence of the later to $I_t^{\theta^*}$ and the asymptotic normality of Δ_t^{N,θ^*} is justified in Proposition 5.1.

On the other hand, we identify

$$\frac{\ddot{\ell}^N[\theta^*]}{N} = -I_t^{N,\theta^*} + \widehat{M}_t^{N,\theta^*},$$

in which

$$\widehat{M}_t^{N,\theta^*} = \frac{1}{N}\int_0^t\int_{\mathbb{R}}\Big(\frac{\ddot{f}_{\theta^*}}{f_{\theta^*}} - \big(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\big)^{\otimes 2}\Big)(x)\ \widetilde{J}_J^N(\mathrm{d}s,\mathrm{d}x),$$

so that $(\widehat{M}_s^{N,\theta^*})_{s\geq 0}$ defines a martingale with values in $\mathcal{M}^d$. The squared Frobenius norm of $\widehat{M}_t^{N,\theta^*}$, denoted $\|\widehat{M}_t^{N,\theta^*}\|_F^2$, can then be bounded in expectation, with the following identity:

$$\mathbb{E}_{\mathbb{P}_N^{\theta^*}}\big(\|\widehat{M}_t^{N,\theta^*}\|_F^2\big) = \frac{1}{N}\int_0^t \mu_s^{N,\theta^*}\Big[\Big\|\Big(\frac{\ddot{f}_{\theta^*}}{f_{\theta^*}} - \big(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\big)^{\otimes 2}\Big)\Big\|_F^2\cdot f_{\theta^*}\Big]\,\mathrm{d}s. \tag{22}$$

which is obtained by expressing the expectation as the sum of the quadratic variations of the individual matrix entries

$$\sum_{k,\ell=1}^d \mathbb{E}_{\mathbb{P}_N^{\theta^*}}\big(\langle\widehat{M}^{N,\theta^*}[k,\ell]\rangle_t\big) = \sum_{k,\ell=1}^d \frac{1}{N}\int_0^t \mu_s^{N,\theta^*}\Big[\Big(\Big(\frac{\ddot{f}_{\theta^*}}{f_{\theta^*}} - \big(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\big)^{\otimes 2}\Big)[k,\ell]\Big)^2\cdot f_{\theta^*}\Big]\,\mathrm{d}s\,.$$

Then, uniformly over $x\in\mathbb{R}$, we have that

$$\Big\|\Big(\frac{\ddot{f}_{\theta^*}}{f_{\theta^*}}(x) - \big(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\big)^{\otimes 2}\Big)(x)\Big\|_F^2 \leq 2\Big(\Big\|\frac{\ddot{f}_{\theta^*}}{f_{\theta^*}}(x)\Big\|_F^2 + \Big\|\Big(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\Big)^{\otimes 2}(x)\Big\|_F^2\Big) \leq B_2^2 + B_1^4 < \infty\,,$$

where $B_2 = \sup_{\theta,x}\Big\|\ddot{f}_\theta(x)/f_\theta(x)\Big\|_F^2$ is finite under Assumptions 4 and 8, which entails that $\mathbb{E}_{\mathbb{P}_N^{\theta^*}}\big(\|\widehat{M}_t^{N,\theta^*}\|_F^2\big)$ tends to 0 as $N\to\infty$. In particular, $\widehat{M}_t^{N,\theta^*}$ tends to 0 in probability. Though the estimate is uniform in θ, $\hat{\theta}_t^N$ depends on the whole behavior of the Poisson processes, so that it cannot directly be applied to the sequence $\hat{\theta}_t^N$ for instance.

On the other hand, we see as in (20) the following upper-bound on the event $\mathcal{J}^N$, that was defined in (8):

$$\frac{\|\ddot{\ell}^N[\theta]-\ddot{\ell}^N[\theta^*]\|_F^2}{N} \leq 4B_0\, t\, \sup_x \Big\|\Big(\frac{\ddot{f}_\theta}{f_\theta}-\frac{\ddot{f}_{\theta^*}}{f_{\theta^*}}\Big)(x)\Big\|_F^2 + \Big\|\Big(\big(\frac{\dot{f}_\theta}{f_\theta}\big)^{\otimes 2}-\big(\frac{\dot{f}_{\theta^*}}{f_{\theta^*}}\big)^{\otimes 2}\Big)(x)\Big\|_F^2 + B_0\, t \sup_x \Big\|\frac{\ddot{f}_\theta-\ddot{f}_{\theta^*}}{f_{\theta^*}}(x)\Big\|_F^2,$$

which tends to 0, as θ tends to θ^*, by the uniform continuity in θ^* of $\ddot{f}_\theta$, $\dot{f}_\theta$ and f_θ. Let

$$\widehat{I}_t^{N,\theta^*} = \frac{-1}{N}\int_0^1 \ddot{\ell}^N[\theta^*+\xi\cdot(\hat{\theta}_t^N-\theta^*)]\,\mathrm{d}\xi\,.$$

Since $\hat{\theta}_t^N$ is a consistent estimator, $\widehat{I}_t^{N,\theta^*}$ is therefore equal to I_t^{N,θ^*} up to a term that converges in $\mathbb{P}_N^{\theta^*}$-probability to 0. So, recalling Proposition 5.1, we deduce that the following convergence in law under $\mathbb{P}_N^{\theta^*}$ holds that

$$(\widehat{I}_t^{N,\theta^*}, \Delta_t^{N,\theta^*}) \xrightarrow{\mathcal{L}} (I_t^{\theta^*}, (I_t^{\theta^*})^{1/2} W_t),$$

as $N\to\infty$. We can then apply the continuous mapping theorem to the function $(x,y)\mapsto -\mathbf{1}_{\{\det(x)\neq 0\}} x^{-1}\, y$ (because $\det(I_t^{\theta^*})\neq 0$ so that the set of discontinuity points is negligible for the limiting distribution). Using (21), we deduce that $N^{-1/2}(\hat{\theta}_t^N-\theta^*)$ converges in distribution to $\mathcal{N}\big(0,(I_t^{\theta^*})^{-1}\big)$, under $\mathbb{P}_N^{\theta^*}$, as $N\to\infty$.

In addition, because $\widehat{I}_t^{N,\theta^*}-I_t^{N,\theta^*}$ converges in $\mathbb{P}_N^{\theta^*}$-probability to 0, we can state the LAN property (7) with $\widehat{I}_t^{N,\theta^*}$ instead of I_t^{N,θ^*}. With (21), [27, Theorem 7.12(ii)] thus proves the local asymptotic minimax bound in Theorem 2.8. □

6.3. **Optimality properties of the maximum likelihood estimator in the absence of large jumps.**

6.3.1. *Proof of the optimality properties of the MLE in the absence of large jumps.*
In this section we show how the results of Theorem 2.9 are deduced from [29, Section III.1] by adapting the approach proposed in [15]. They rely on the next two estimates in combination with the LAN property.

Let us denote $\lambda_{\theta^*}^N[h] = \ell^N[\theta^*+\frac{h}{\sqrt{N}}] = \log L_t^{(\theta^*+h/\sqrt{N})/\theta^*}$, for any $h\in\mathbb{R}^d$ such that $\theta^*+h/\sqrt{N}\in\Theta$.

Lemma 6.1. *Under Assumptions 2 and 4, for any $r\geq d$, there exists $C>0$ such that the following holds for any h,h and any $\theta^*\in\Theta$:*

$$\mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\Big|\exp\Big(\frac{\lambda_{\theta^*}^N[h]}{r}\Big)-\exp\Big(\frac{\lambda_{\theta^*}^N[\mathrm{h}]}{r}\Big)\Big|^r\Big] \leq C\cdot\|h-\mathrm{h}\|^r\,.$$

Lemma 6.2. *Suppose $\varphi\equiv 0$ and Assumptions 1 to 7 and 9. Then, for any $r>0$, there exists $C>0$ such that the following holds for any $\theta^*\in\Theta$ and any h such that $\theta^*+\frac{h}{\sqrt{N}}\in\Theta$:*

$$\mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\exp\Big(\frac{\lambda_{\theta^*}^N[h]}{2}\Big)\Big] \leq C\,\|h\|^{-r}\,. \tag{23}$$

At the moment, we can only proceed with (23) in the case where $\varphi\equiv 0$, which is thus assumed in this Section 6.3.1. This assumption appears typically needed for the following quantitative result of propagation of chaos.

Lemma 6.3. *Suppose $\varphi \equiv 0$ and Assumptions 2 to 4 and 9. For any $r > 2$, the following holds for some $C > 0$ that only depends on r, t and $\mathbb{E}[|X_0^1|^{2r}]$*

$$\sup_{s\in[0,t]} \mathbb{E}_{\mathbb{P}^N_{\theta^*}}[\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar{\mu}_s^{\theta^*})^r] \leq \frac{C}{N^{r/2}}.$$

Let $(\bar{X}^{i,N,\theta^*}_{[0,t]})_{i\in[\![1,N]\!]}$ be a family of replicates of the Mc Kean-Vlasov dynamics, with initial condition specified by $\bar{X}_0^{i,N,\theta^*} = X_0^{i,N,\theta^*} = X_0^i$, and for any $s \in [0,t]$:

$$\bar{\mu}_s^{N,\theta^*}(\mathrm{d}x) := \frac{1}{N}\sum_{i=1}^N \delta_{\bar{X}_s^{i,N,\theta^*}}(\mathrm{d}x)\,.$$

Remark 6.4. *The idea is to exploit a diagonal coupling with the same Poisson measure for the j-th particle and control together $|X_s^{i,N,\theta^*} - \bar{X}_s^{i,N,\theta^*}|^r$ and $\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar{\mu}_s^{N,\theta^*})^r$ by means of the Grönwall Lemma. More precisely, the Wasserstein norm appears as the natural quantity to bound the predictible part of the interaction term, together with a correction term which we handle with concentration inequalities taken from [20], while the martingale terms are controlled by the Burckholder-Davis-Gundy inequality. This last control fails to produce satisfying bounds if $\varphi \neq 0$, due to the dependency of the jump rate on the state of the particle. We conjecture that such a strong result actually does not hold in this case where $\varphi \neq 0$.*

Proof of Lemma 6.1. We start the proof by noticing that

$$\begin{aligned}\mathbb{E}_{\mathbb{P}^N_{\theta^*}}\Big[\Big|\exp\Big(\frac{\lambda^N_{\theta^*}[h]}{r}\Big) - \exp\Big(\frac{\lambda^N_{\theta^*}[\mathrm{h}]}{r}\Big)\Big|^r\Big] &= \mathbb{E}_{\mathbb{P}^N_{\theta^*}}\Big[\|h-\mathrm{h}\|^r \cdot \Big\|\int_0^1 \partial_h \exp\Big(\frac{\lambda^N_{\theta^*}[.]}{r}\Big)[h_\xi]\,\mathrm{d}\xi\Big\|^r\Big] \\ &\leq \|h-\mathrm{h}\|^r \int_0^1 \mathbb{E}_{\mathbb{P}^N_{\theta^*}}\Big[\Big\|\partial_h \exp\Big(\frac{\lambda^N_{\theta^*}[.]}{r}\Big)[h_\xi]\Big\|^r\Big]\,\mathrm{d}\xi,\end{aligned} \tag{24}$$

thanks to the Jensen inequality and the Fubini theorem, where $h_\xi = h + \xi\,(\mathrm{h} - h)$. Using the chain rule and the relation between λ^N and ℓ^N we get

$$\Big\|\partial_h \exp\Big(\frac{\lambda^N_{\theta^*}[.]}{r}\Big)[h_\xi]\Big\|^r = \exp\Big(\ell^N[\theta_\xi]\Big)\cdot\Big(\frac{1}{r\sqrt{N}}\Big)^r \|\partial_\theta \ell^N[\theta_\xi]\|^r\,,$$

where $\theta_\xi = \theta^* + \frac{h_\xi}{\sqrt{N}}$. For θ, we recall also that $\exp\circ\ell^N[\theta] = L_t^{\theta/\theta^*}$ is the density of $\mathbb{P}^\theta_N$ with respect to $\mathbb{P}^{\theta^*}_N$ on the time-interval $[0,t]$, so that for any $\xi \in [0,1]$:

$$\mathbb{E}_{\mathbb{P}^{\theta^*}_N}\Big[\Big\|\partial_h \exp\Big(\frac{\lambda^N_{\theta^*}[.]}{r}\Big)[h_\xi]\Big\|^r\Big] = \Big(\frac{1}{r\sqrt{N}}\Big)^r \mathbb{E}_{\mathbb{P}^{\theta_\xi}_N}\Big[\|\partial_\theta \ell^N[\theta_\xi]\|^r\Big]\,. \tag{25}$$

Under $\mathbb{P}^\theta_N$, using (8) and (9), we identify the law of $\partial_\theta \ell^N[\theta]$ from (10) as the one of:

$$\int_{[0,t]\times\mathbb{R}} \frac{\dot{f}_\theta(x)}{f_\theta(x)}\, J^{N,\theta}(\mathrm{d}s,\mathrm{d}x) - \int_0^t \frac{\dot{f}_\theta(x)}{f_{\theta^*}(x)}\, \Upsilon^{N,\theta/\theta^*}(\mathrm{d}s,\mathrm{d}x)\,,$$

where $\Upsilon^{N,\theta/\theta^*}(\mathrm{d}s,\mathrm{d}x) = N\, f_{\theta^*}(x)\, \mu_s^{N,\theta}(\mathrm{d}x)\,\mathrm{d}s$. Therefore $N^{-1/2}\partial_\theta \ell^N[\theta]$ has the law of $\Delta_t^{N,\theta}$ where $(\Delta_s^{N,\theta})_{s\in[0,t]}$ is a martingale defined as follows, in analogy with (13),

$$\begin{aligned}\Delta_t^{N,\theta} &= \frac{1}{\sqrt{N}} \iint_{[0,t]\times\mathbb{R}} \frac{\dot f_\theta}{f_\theta}(x)\ \widetilde{J}^{N,\theta}(\mathrm{d}s,\mathrm{d}x)\\ &= \frac{1}{\sqrt{N}} \sum_{j=1}^N \iint_{[0,t]\times\mathbb{R}^+} \frac{\dot f_\theta}{f_\theta}\big(X_{s-}^{j,N,\theta}\big)\mathbf{1}_{\left\{z\le f_\theta\left(X_{s-}^{j,N,\theta}\right)\right\}}\widetilde{\bar\pi}^j(\mathrm{d}s,\mathrm{d}z)\,.\end{aligned}$$

We can thus apply the Burkholder-Davis-Gundy inequality to the martingale $\Delta_t^{N,\theta}$ with $r\ge 2$, see [31], Theorem 20.12, then the Jensen inequality, and deduce that:

$$\begin{aligned}&\mathbb{E}_{\mathbb{P}_N^\theta}\Big[|N^{-1/2}\partial_\theta\ell^N[\theta]|^r\Big] = \mathbb{E}_{\mathbb{P}_N^\theta}\Big[|\Delta_t^{N,\theta}|^r\Big]\\ &\le \mathbb{E}_{\mathbb{P}_N^\theta}\Big[[\Delta_t^{N,\theta}]^{r/2}\Big] = \mathbb{E}_{\mathbb{P}_N^\theta}\Big[\Big|\frac{1}{N}\sum_{j=1}^N \iint_{[0,t]\times\mathbb{R}^+}\Big(\frac{\dot f_\theta}{f_\theta}\big(X_{s-}^{j,N,\theta}\big)\Big)^2\mathbf{1}_{\left\{z\le f_\theta\left(X_{s-}^{j,N,\theta}\right)\right\}}\bar\pi^j(\mathrm{d}s,\mathrm{d}z)\Big|^{r/2}\Big]\\ &\le B_1^r\,\mathbb{E}_{\mathbb{P}_N^\theta}\Big[(\bar\pi^1([0,t]\times[0,B_0])^{r/2}\Big] = C(t,B_0,B_1)<\infty.\end{aligned}$$

Recalling in addition (24) and (25), we conclude Lemma 6.1 with $C = r^{-r}\cdot C_M^{r/2}$. □

Proof of Lemma 6.3. Recall that $(\bar X_{[0,t]}^{i,N,\theta^*})_{i\in[\![1,N]\!]}$ denotes a family of replicates of the Mc Kean-Vlasov dynamics, with initial condition specified by $\bar X_0^{i,N,\theta^*} = X_0^{i,N,\theta^*} = X_0^{i,\theta^*}$, and for any $s\in[0,t]$:

$$\bar\mu_s^{N,\theta^*}(\mathrm{d}x) := \frac{1}{N}\sum_{i=1}^N \delta_{\bar X_s^{i,N,\theta^*}}(\mathrm{d}x)\,.$$

Since

$$\mathcal{W}_1(\mu_s^{N,\theta^*},\bar\mu_s^{\theta^*}) \le \mathcal{W}_1(\mu_s^{N,\theta^*},\bar\mu_s^{N,\theta^*}) + \mathcal{W}_1(\bar\mu_s^{N,\theta^*},\bar\mu_s^{\theta^*}),$$

it holds that

$$2^{-r+1}\mathbb{E}_{\mathbb{P}_{\theta^*}^N}[\mathcal{W}_1(\mu_s^{N,\theta^*},\bar\mu_s^{\theta^*})^r] \le \mathbb{E}_{\mathbb{P}_{\theta^*}^N}[\mathcal{W}_1(\mu_s^{N,\theta^*},\bar\mu_s^{N,\theta^*})^r] + \mathbb{E}_{\mathbb{P}_{\theta^*}^N}[\mathcal{W}_1(\bar\mu_s^{N,\theta^*},\bar\mu_s^{\theta^*})^r]\,. \tag{26}$$

Recall that $(\bar X_{[0,t]}^{i,N,\theta^*})_{i\in[\![1,N]\!]}$ follows the Mc-Kean Vlasov dynamics driven by $\bar\mu_{[0,t]}^{\theta^*}$ while $\bar X_0^{i,N,\theta^*}$ are i.i.d. samples with distribution $\bar\mu_0^{\theta^*}$. Thus, for any $s\in[0,t]$, $\bar X_s^{i,N,\theta^*}$ are i.i.d. samples with distribution $\bar\mu_s^{\theta^*}$.

Let us denote by $M_r(\theta^*)$ the upper-bound on the moment of order $2r$ of $\bar\mu_s^{\theta^*}$:

$$M_r(\theta^*) := \mathbb{E}\Big[\sup_{s\le t}|\bar X_s^{1,N,\theta^*}|^{2r}\Big].$$

Since

$$\begin{aligned}\sup_{s\le t}|\bar X_s^{1,N,\theta^*}|^{2r} \le \big(2^{2r-1}+(16t)^{2r-1}\|b\|_{Lip}^{2r}\big)\,|X_0^1|^{2r} + (8t)^{2r}\|b\|_{Lip}^{2r}\int_0^T \sup_{u\le s}|\bar X_u^{1,N,\theta^*}|^{2r}\mathrm{d}s\\ + t^{2r}8^{2r-1}\big(|b(X_0^1)|^{2r} + |m|\sup_{\theta^*\in\Theta}\|f_{\theta^*}\|_\infty)^{2r}\big).\end{aligned}$$

Using Grönwall lemma, for all $\theta^*\in\Theta$ we deduce that

$$M_r(\theta^*) \le C\Big(t,r,m,\sup_{\theta\in\Theta}\|f_\theta\|_\infty,\|b\|_{Lip},\mathbb{E}\Big[|X_0^1|^{2r}\Big]\Big) < \infty \tag{27}$$

and since the previous bound doesn't depend on θ^*, it is uniform: $\sup_{\theta^*\in\Theta} M_r(\theta^*) \le C$.

We can therefore exploit [20, Theorem 2] and deduce that there exists C_c depending only on r and M_r such that

$$\sup_{s\in[0,t]} \mathbb{E}_{\mathbb{P}^N_{\theta^*}}[\mathcal{W}_1(\bar\mu_s^{N,\theta^*}, \bar\mu_s^{\theta^*})^r] \le \frac{C_c}{N^{r/2}}\,. \tag{28}$$

Our aim now is to show a similar bound for $\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar\mu_s^{N,\theta^*})$ in (26). Using the diagonal coupling and the Jensen inequality, we deduce that

$$\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar\mu_s^{N,\theta^*}) \le \frac{1}{N}\sum_{i\le N} |\bar X_s^{i,N,\theta^*} - X_s^{i,N,\theta^*}|,$$

and

$$\mathbb{E}_{\mathbb{P}^N_{\theta^*}}[\mathcal{W}_1(\mu_s^{N,\theta^*}, \bar\mu_s^{N,\theta^*})^r] \le \frac{1}{N}\sum_{i\le N} \mathbb{E}_{\mathbb{P}^N_{\theta^*}}[|X_s^{i,N,\theta^*} - \bar X_s^{i,N,\theta^*}|^r]\,. \tag{29}$$

There exists $C_d > 0$ depending only on r and t such that for any $i \in [\![1, N]\!]$

$$\frac{1}{C_d}|X_s^{i,N,\theta^*} - \bar X_s^{i,N,\theta^*}|^r \le \int_0^s (\mathcal{B}_v^{i,N,\theta^*})^r \mathrm{d}v + (\mathcal{D}_s^{i,N,\theta^*})^r + |\mathcal{N}_s^{i,N,\theta^*}|^r\,,$$

where

$$\mathcal{B}_s^{i,N,\theta^*} = |b(X_s^{i,N,\theta^*}) - b(\bar X_s^{i,N,\theta^*})| + |\sum_{j=1}^N \int_{\mathbb{R}_+} \frac{u}{N} f_{\theta^*}(X_s^{j,N,\theta^*})\nu(\mathrm{d}u) - m\ \bar\mu_s^{\theta^*}[f_{\theta^*}]|,$$

so that

$$2^{-r+1}(\mathcal{B}_s^{i,N,\theta^*})^r \le \|b\|_{Lip}^r \cdot |X_s^{i,N,\theta^*} - \bar X_s^{i,N,\theta^*}|^r + m^r\, \|f_{\theta^*}\|_{Lip}^r\ W_1(\mu_s^{N,\theta^*}, \bar\mu_s^{\theta^*})^r\,, \tag{30}$$

with $\|.\|_{Lip}$ denoting the Lipschitz norm, while

$$\mathcal{D}_s^{i,N,\theta^*} = \Big|\int_0^s \int_{\mathbb{R}} \frac{u}{N} f_{\theta^*}(X_v^{i,N,\theta^*})\nu(\mathrm{d}u)\,\mathrm{d}v\Big| \le \frac{m\,s\,B_0}{N} \tag{31}$$

and $\mathcal{N}_{[0,t]}^{i,N,\theta^*}$ is a martingale starting from 0 defined, for any $s \in [0,t]$, as follows

$$\mathrm{d}\mathcal{N}_s^{i,N,\theta^*} = \sum_{j\neq i} \int_{\mathbb{R}_+}\int_{\mathbb{R}} \frac{u}{N}\mathbf{1}_{\{z\le f_{\theta^*}(X_{s-}^{j,N,\theta^*})\}}\widetilde{\pi}^j(\mathrm{d}s,\mathrm{d}z,\mathrm{d}u)\,.$$

Thanks to the Burkholder-Davis-Gundy inequality see [31], Theorem 20.12, then the Jensen inequality,

(32)

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}^N_{\theta^*}}\Big[\sup_{s\in[0,t]}|\mathcal{N}^{i,N,\theta^*}_s|^r\Big] &\le \mathbb{E}_{\mathbb{P}^N_{\theta^*}}\left[\left(\frac{1}{N}\sum_{j\neq i}\int_{[0,t]}\int_{\mathbb{R}_+}\int_{\mathbb{R}}\frac{u^2}{\sqrt{N}}\mathbf{1}_{\left\{z\le f_{\theta^*}(X^{j,N,\theta^*}_{s-})\right\}}\pi^j(\mathrm{d}s,\mathrm{d}z,\mathrm{d}u)\right)^{r/2}\right]\\
&\le N^{-r/2}\,\mathbb{E}_{\mathbb{P}^N_{\theta^*}}\left[\frac{1}{N}\sum_{j\neq i}\left(\int_{[0,t]}\int_{\mathbb{R}_+}\int_{\mathbb{R}}u^2\mathbf{1}_{\left\{z\le f_{\theta^*}(X^{j,N,\theta^*}_{s-})\right\}}\pi^j(\mathrm{d}s,\mathrm{d}z,\mathrm{d}u)\right)^{r/2}\right]\\
&\le N^{-r/2}\,\mathbb{E}_{\mathbb{P}^N_{\theta^*}}\left[\left(\int_{[0,t]}\int_{\mathbb{R}_+}\int_{\mathbb{R}}u^2\mathbf{1}_{\left\{z\le f_{\theta^*}(X^{j,N,\theta^*}_{s-})\right\}}\pi^1(\mathrm{d}s,\mathrm{d}z,\mathrm{d}u)\right)^{r/2}\right]\\
&\le N^{-r/2}\,\mathbb{E}_{\mathbb{P}^N_{\theta^*}}\left[\left(\int_{[0,t]}\int_{[0,B_0]}\int_{\mathbb{R}}u^2\pi^1(\mathrm{d}s,\mathrm{d}z,\mathrm{d}u)\right)^{r/2}\right]\le CN^{-r/2},
\end{aligned}$$

as the compound Poisson process in the last line admits the moments of any order under Assumption 9 (ii).

We can thus justify a Grönwall approach on the quantity

$$\mathcal{G}^{N,\theta^*}_s=\frac{1}{N}\sum_{i\le N}\mathbb{E}_{\mathbb{P}^N_{\theta^*}}[|\bar{X}^{i,N,\theta^*}_s-X^{i,N,\theta^*}_s|^r],$$

as s varies from 0 to t. The Wasserstein term in (30) is to be upper-bounded as in our initial steps, by recalling (26), (29) and (28). The martingale term is to be treated thanks to (32) together with the estimates given in (31) and the one associated with (28). The leading error term is the one of the martingale, of order $N^{-r/2}$. Putting all these estimates together, for any sufficiently large N, we arrive at the following upper-bound:

$$\mathcal{G}^{N,\theta^*}_s\le\frac{2C_d\,C_{\mathcal{N}}}{N^{r/2}}+C_d\,C_{\mathcal{G}}\int_0^s\mathcal{G}^{N,\theta^*}_v\,\mathrm{d}v\,,$$

where $C_{\mathcal{G}}=\|b\|^r_{Lip}+4^{r-1}m^r\,B^r_L$. The Grönwall Lemma implies that there exists $C>0$ such that $\sup_{s\in[0,t]}\mathcal{G}^{N,\theta^*}_s\le C/N^{r/2}$. The proof of Lemma 6.3 is finally concluded with (26) and (28). □

We can next proceed with the proof of (23).

Proof of Lemma 6.2. Let $\theta^*\in\Theta$ be fixed and $h\in\mathbb{R}^d$ be such that $\theta=\theta^*+h/\sqrt{N}\in\Theta$. Recall the notation $\lambda^N_{\theta^*}[h]=\ell^N[\theta^*+h/\sqrt{N}]=\log L_t^{(\theta^*+h/\sqrt{N})/\theta^*}$.

From (5) we know that

$$\ell^N[\theta]=P_t^{N,\theta/\theta^*}-R_t^{N,\theta/\theta^*},$$

where, for any $t\ge0$,

$$P_t^{N,\theta/\theta^*}=\sum_{j=1}^N\iint_{[0,t]\times\mathbb{R}^+}\log\Big(\frac{f_\theta}{f_{\theta^*}}\Big)\big(X^{j,N,\theta^*}_{s-}\big)\mathbf{1}_{\left\{z\le f_{\theta^*}\left(X^{j,N,\theta^*}_{s-}\right)\right\}}\bar{\pi}^j(\mathrm{d}s,\mathrm{d}z)$$

is the Poissonian term, while

$$R_t^{N,\theta/\theta^*} = N \int_{[0,t]} \int_{\mathbb{R}^+} (f_\theta - f_{\theta^*})\big(x\big)\, \mu_s^{N,\theta^*}(\mathrm{d}x)\, \mathrm{d}s$$
$$= \sum_{j=1}^{N} \int_0^t (f_\theta - f_{\theta^*})\big(X_s^{j,N,\theta^*}\big)\, \mathrm{d}s$$

refers to the jump rate. As we want to use the Hölder inequality to combine the term proportional to $P_t^{N,\theta/\theta^*}$ with a compensation that makes the exponential a martingale, we introduce the following quantity:

$$G_t^{N,\theta/\theta^*} = \frac{N}{3} \int_{[0,t]} \int_{\mathbb{R}^+} \Big[\Big(\frac{f_\theta}{f_{\theta^*}}\Big)^{3/4} - 1\Big](x)\, f_{\theta^*}(x)\, \mu_s^{N,\theta^*}(\mathrm{d}x)\, \mathrm{d}s$$
$$= \sum_{j=1}^{N} \frac{1}{3} \int_{[0,t]} \Big[\Big(\frac{f_\theta}{f_{\theta^*}}\Big)^{3/4} - 1\Big](X_s^{j,N,\theta^*})\, f_{\theta^*}(X_s^{j,N,\theta^*})\, \mathrm{d}s\,.$$

It is chosen such that

$$Z_t^{N,\theta/\theta^*} = \exp\Big[\frac{3}{4} P_t^{N,\theta/\theta^*} - 3 G_t^{N,\theta/\theta^*}\Big]$$

defines an exponential local martingale under $\mathbb{P}_{\theta^*}^N$, and thus a supermartingale, in line with [6, Section 5.2].Therefore

$$\mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[Z_t^{N,\theta/\theta^*}\Big] \le 1.$$

By the Hölder inequality

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\exp\Big(\frac{\ell^N[\theta]}{2}\Big)\Big] &\le \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[Z_t^{N,\theta/\theta^*}\Big]^{2/3} \cdot \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\exp\Big(\mathcal{K}_t^{N,\theta/\theta^*}\Big)\Big]^{1/3} \\ &\le \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\exp\Big(\mathcal{K}_t^{N,\theta/\theta^*}\Big)\Big]^{1/3}, \end{aligned} \tag{33}$$

where

$$\begin{aligned} \mathcal{K}_t^{N,\theta/\theta^*} &= 3 \times 2\, G_t^{N,\theta/\theta^*} - \frac{3}{2} R_t^{N,\theta/\theta^*} \\ &= N \int_0^t \int_{\mathbb{R}^+} \mathrm{k}\Big(\frac{f_\theta}{f_{\theta^*}}\Big)(x)\, f_{\theta^*}(x)\, \mu_s^{N,\theta^*}(\mathrm{d}x)\, \mathrm{d}s\,, \end{aligned} \tag{34}$$

and $\mathrm{k}(r) = 2r^{3/4} - \frac{3}{2}r - \frac{1}{2}$. Note that $\mathrm{k}(1) = \mathrm{k}'(1) = 0$, $\mathrm{k}''(1) = \frac{-3}{8}$. k is also non-positive, hence

$$\exp\Big(\mathcal{K}_t^{N,\theta/\theta^*}\Big) \le 1\,. \tag{35}$$

Note also that

$$\mathrm{k}(r) = 0 \iff r = 1,$$

and $\mathrm{k}''(r) \le 0$, therefore k is concave.

Due to the second order Taylor expansion of k around 1, let $\varepsilon_{\mathrm{k}} > 0$ be such that $\mathrm{k}(1+r) \le \frac{-3r^2}{20}$ for any $|r| \le \varepsilon_{\mathrm{k}}$. Thanks to Assumption 5, then Assumption 4 for the boundedness of $\dot{f}_{\theta^*}/f_{\theta^*}$, we deduce that there exists $\hat{C}, \hat{\varepsilon}_{\mathrm{k}} > 0$ such that the following holds for any $\theta \in B(\theta^*, \hat{\varepsilon}_{\mathrm{k}})$ and any $x \in \mathbb{R}_+$:

$$\mathrm{k}\Big(\frac{f_\theta}{f_{\theta^*}}\Big)(x) \le -\frac{3}{20}\Big((\theta - \theta^*)^\intercal \frac{\dot{f}_{\theta^*}}{f_{\theta^*}}(x)\Big)^2 + \hat{C} \cdot \|\theta - \theta^*\|^3\,.$$

Thus, for any such θ:

$$\mathcal{K}_t^{N,\theta/\theta^*} \leq -\frac{3N}{20}(\theta-\theta^*)^\mathsf{T} I_t^{N,\theta^*}(\theta-\theta^*) + N\,t\,\hat{C}\,\|\theta-\theta^*\|^3\,, \tag{36}$$

where I_t^{N,θ^*} is defined in (15). Note that for a fixed value of h, using Proposition 5.1, with $I_t^{\theta^*}$ given by (6), we deduce the following limiting upper-bound:

$$\limsup_{N\to\infty} \mathcal{K}_t^{N,(\theta^*+h/\sqrt{N})/\theta^*} \leq -\frac{3}{20}\,h^\mathsf{T} I_t^{\theta^*} h\,.$$

Let

$$\mathcal{H}_{loc}^N = \Big\{\mathcal{K}_t^{N,(\theta^*+h/\sqrt{N})/\theta^*} \leq -\frac{1}{10} h^\mathsf{T} I_t^{\theta^*} h\Big\}.$$

On the complementary of the event $\mathcal{H}_{loc}^N$, because of (36):

$$6h^\mathsf{T} I_t^{N,\theta^*} h < 4h^\mathsf{T} I_t^{\theta^*} h + 40t\,\hat{C}\,\hat{\varepsilon}_k\,\|h\|^2,$$

By possibly restricting the size of $\hat{\varepsilon}_k$ (independently of h), since $(I_t^\theta)_\theta$ is uniformly non-degenerated as noted in Remark 2.2, we may assume that the last term in the r.h.s. is smaller than $h^\mathsf{T} I_t^{\theta^*} h$. So we deduce that the complementary of $\mathcal{H}_{loc}^N$ is included in the following rare event:

$$\mathcal{I}^{N,\theta^*} = \Big\{h^\mathsf{T} I_t^{N,\theta^*} h \leq \frac{5}{6} h^\mathsf{T} I_t^{\theta^*} h\Big\}\,.$$

Using Remark 2.2, $(I_t^\theta)_\theta$ is uniformly non-degenerated, therefore we can write, for some positives constants c, C, C_0,

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\exp\Big(\mathcal{K}_t^{N,\theta/\theta^*}\Big)\Big] &\leq \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\exp\Big(\mathcal{K}_t^{N,\theta/\theta^*}\Big)\mathbf{1}_{\mathcal{H}_{loc}^N}\Big] + \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\Big[\mathbf{1}_{(\mathcal{H}_{loc}^N)^c}\Big] \\ &\leq \exp\Big(-\frac{1}{10}\,h^\mathsf{T} I_t^{\theta^*} h\Big) + \mathbb{P}_{\theta^*}^N(\mathcal{I}^{N,\theta^*}) \leq \exp(-c\|h\|^2) + \mathbb{P}_{\theta^*}^N(\mathcal{I}^{N,\theta^*}) \leq C\,\|h\|^{-r} + C_0\,N^{-r/2}\,.\end{aligned} \tag{37}$$

Indeed, under Assumption 7, using the fact that the moment of order $2r$ of $\bar{\mu}_s^{\theta^*}$ is upper-bounded (see (27)) uniformly in θ^* and $s\in[0,t]$, as a consequence of Lemma 6.3 we can write

$$\begin{aligned}\mathbb{P}_{\theta^*}^N(\mathcal{I}^{N,\theta^*}) &\leq \mathbb{P}_{\theta^*}^N(\|I_t^{N,\theta^*} - I_t^{\theta^*}\| \geq C) \\ &\leq \mathbb{P}_{\theta^*}^N\Big(\int_0^t \|\mu_s^{N,\theta^*}\Big[\frac{(\dot{f}_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\Big] - \bar{\mu}_s^{\theta^*}\Big[\frac{(\dot{f}_{\theta^*})^{\otimes 2}}{f_{\theta^*}}\Big]\|\,\mathrm{d}s \geq C\Big) \\ &\leq \mathbb{P}_{\theta^*}^N\Big(B_1^2\int_0^t \mathcal{W}_1(\mu_s^{N,\theta^*},\bar{\mu}_s^{\theta^*})\,\mathrm{d}s \geq C\Big) \\ &\leq C(t,r)\sup_{s\in[0,t]}\mathbb{E}_{\mathbb{P}_{\theta^*}^N}[\mathcal{W}_1(\mu_s^{N,\theta^*},\bar{\mu}_s^{\theta^*})^r] \leq \frac{C(t,r)}{N^{r/2}}\,.\end{aligned} \tag{38}$$

As a consequence of (33), (37) and (38), for any $h\in B(0,\hat{\varepsilon}_\mathrm{k}\sqrt{N})$, the claim of Lemma 6.2 follows.

We consider now the case of $\theta\notin B(\theta^*,\hat{\varepsilon}_\mathrm{k})$. Due to Assumption 7, Remark 2.6, the fact that $k(r)<0$ for any $r\neq 1$, and $f_{\theta^*}>0$, the following quantity is strictly positive:

$$\mathfrak{K}_t^{\theta/\theta^*} = -\int_0^t\int_{\mathbb{R}^+} \mathrm{k}\Big(\frac{f_\theta}{f_{\theta^*}}\Big)(x)\,f_{\theta^*}(x)\,\bar{\mu}_s^{\theta^*}(\mathrm{d}x)\,\mathrm{d}s > 0.$$

Moreover, by continuity of $\mathfrak{K}_t^{\theta/\theta^*}$ with respect with (θ, θ^*), there is a positive uniform lower-bound

$$\inf_{\theta,\theta^*:\|\theta-\theta^*\|>\hat{\varepsilon}_{\mathrm{k}}} \mathfrak{K}_t^{\theta/\theta^*} > 0.$$

Let

$$\mathcal{H}_{gl}^{N,\theta/\theta^*} = \Big\{ \mathcal{K}_t^{N,\theta/\theta^*} \leq -\frac{N}{2}\mathfrak{K}_t^{\theta/\theta^*} \Big\},$$

where $\mathcal{K}_t^{N,\theta/\theta^*}$ is defined by (34). The complementary of $\mathcal{H}_{gl}^{N,\theta/\theta^*}$ is included in the following rare event:

$$\mathfrak{J}^{N,\theta/\theta^*} = \Big\{ -\int_0^t \int_{\mathbb{R}^+} \mathrm{k}\Big(\frac{f_\theta}{f_{\theta^*}}\Big)(x)\, f_{\theta^*}(x)\, \mu_s^{N,\theta^*}(\mathrm{d}x)\, \mathrm{d}s \leq \frac{1}{2}\mathfrak{K}_t^{\theta/\theta^*} \Big\}.$$

Indeed, we can control its probability with the help of Lemma 6.3 as follows:

$$\begin{aligned} \mathbb{P}_{\theta^*}^N \Big(\mathfrak{J}^{N,\theta/\theta^*}\Big) &\leq \mathbb{P}_{\theta^*}^N \left(\int_0^t \int_{\mathbb{R}^+} \mathrm{k}\Big(\frac{f_\theta}{f_{\theta^*}}\Big)(x)\, f_{\theta^*}(x) \left[\mu_s^{N,\theta^*} - \bar{\mu}_s^{\theta^*}\right](\mathrm{d}x)\, \mathrm{d}s \geq \frac{1}{2}\mathfrak{K}_t^{\theta/\theta^*} \right) \\ &\leq \mathbb{P}_{\theta^*}^N \left(\int_0^t \mathcal{W}_1(\mu_s^{N,\theta^*}, \bar{\mu}_s^{\theta^*})\mathrm{d}s > C(t) \right) \leq C(t,r) N^{-r/2}, \end{aligned} \tag{39}$$

where in the last inequality we have used Jensen inequality and Lemma 6.3, and we denoted by $C(t)$ and $C(t,r)$ two generic positive constants depending respectively on t, and on t and r. Now the result easily follows from (35) and (39):

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\left[\exp\left(\mathcal{K}_t^{N,\theta/\theta^*}\right)\right] &\leq \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\left[\exp\left(\mathcal{K}_t^{N,\theta/\theta^*}\right)\mathbf{1}_{\left\{H_{gl}^{N,\theta/\theta^*}\right\}}\right] + \mathbb{E}_{\mathbb{P}_{\theta^*}^N}\left[(\mathcal{H}_{gl}^{N,\theta/\theta^*})^c\right] \\ &\leq \exp(-\frac{N}{2}\mathfrak{K}_t^{\theta/\theta^*}) + \mathbb{P}_{\theta^*}^N\left(\mathfrak{J}^{N,\theta/\theta^*}\right) \leq C(t,r)N^{-r/2}. \end{aligned}$$

Recall that $\theta^* \in \theta$ and $\theta \in \Theta$ are fixed such that $\theta = \theta^* + h/\sqrt{N}$ and $\|\theta - \theta^*\| \geq \hat{\varepsilon}_{\mathrm{k}}$. Therefore from the last inequality we obtain that for all $r \geq 1$,

$$\mathbb{E}_{\mathbb{P}_{\theta^*}^N}\left[\exp\left(\mathcal{K}_t^{N,\theta/\theta^*}\right)\right] \leq C(t,r)\|h\|^{-r/2},$$

which completes the proof of Lemma 6.2. □

Proof of Theorem 2.9. The uniform LAN property given in Theorem 2.9 is Condition N1 of [29, Chapter III]. The non-degeneracy assumption, that $I_t^\theta > 0$ for any θ, corresponds to Condition N2 of [29]. Lemmas 6.1 and 6.2 are respectively Condition N3 and N4 of [29]. The conditions of [29, Theorem III.1.1] are thus satisfied, which entails statement (i) of Theorem 2.9. Statement (ii) is a consequence of [29, Corollary III.1.1] while Statement (iii) is a consequence of [29, Theorem III.1.3]. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. The authors used Mistral AI (Medium 3.5, Version 26.04) for language polishing (grammar and style), drafting initial versions of simulations and result presentations, and refining figure editing. Qwen's and ChatGPT's performance were tested, for instance by asking for references and by extending the draft from the one-dimensional to the multivariate setting (which revealed efficient with the former). All scientific content, analysis, and conclusions were developed independently by the authors. The authors take full responsibility for the content of the published article.

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Appendix A. Technical Details of Numerical Simulations

A.1. **Numerical Methods.** The numerical implementation closely follows the model described in Section 2.1, with discretization and algorithmic choices tailored to ensure stability and efficiency for finite-N systems. Below, we detail the key numerical procedures used in the simulations and statistical analyses.

A.1.1. *Time Discretization and Spike Generation.* The system of SDEs (1) is approximated using a discrete-time scheme according to the parameters summarized in Table 3. The deterministic drift $b(x) = -\alpha x$ is handled via an explicit resolution at each discrete step, while spikes are generated using the Sellke construction [35], a standard method for simulating inhomogeneous Poisson processes. Specifically, for each neuron i and time step k, the spike probability is determined by the instantaneous firing rate $f_\theta(X^{(i,N,\theta)}_{k\Delta t})\Delta t$. These values are deducted iteratively for neuron i from an initial value E^i_0 sampled (independently for each neuron) according to a standard exponential distribution, until negative values are reached. This event corresponds to the spike of the neuron, at which time a new sample of E^i is generated according to the exponential distribution. The construction is such that the remaining value of E^i at each time step remains exponentially distributed, ensuring the desired jump rate while keeping the procedure of random generation at the same amount than the number of jumps plus the number of neurons without any spike.

Symbol	Value	Description
Model Parameters		
N	100	Number of neurons.
α	1.5	Drift coefficient in $b(x) = -\alpha x$.
$\varphi(x)$	$-x$	Reset function, applied to a neuron's state after it spikes.
θ_0^*	2.5	Saturation level of the firing rate function.
θ_1^*	1.0	Inflection point of the firing rate function.
θ_2^*	3.0	Slope at inflection point of the firing rate function.
ν	$\mathcal{U}(\nu_1 - \nu_2, \nu_1 + \nu_2)$	Distribution of jump sizes.
ν_1	2.4	Base effect of a spike on other neurons.
ν_2	$0.8 = \nu_1/3$	dispersion of the effect.
Simulation Settings		
T	10.0 s	Final observation time.
Δt	0.002 s	Time step size.
N_{steps}	5000	Number of time steps ($T/\Delta t$).
X_0^i	$\mathcal{U}(0, 4)$	Initial state distribution for each neuron.
Optimization Parameters		
θ_{init}	$[1.2, 0.8, 4.0]$	Initial guess for MLE optimization.
maxiter	200	Maximum iterations for the L-BFGS-B optimizer.
Bounds	$\theta_0 > 10^{-3}, \theta_2 > 10^{-3}$	Constraints for MLE optimization.

Table 3. Parameter values used in the numerical simulations. The sigmoidal firing rate function is defined as $f_\theta(x) = \theta_0/[1 + \exp(-\theta_2(x - \theta_1))]$, where $\theta = (\theta_0, \theta_1, \theta_2)$.

A.1.2. *Empirical Fisher Information.* The empirical Fisher information matrix I_t^{N,θ^*} is computed according to (15), with the discretized in time version of the empirical measure process μ^{N,θ^*} (Equation 2), and the analytically computed gradient $\dot{f}_{\theta^*}$. The quantity Δ_t^{N,θ^*} which appears in the LAN property (Equation 7) is similarly computed according to (13), by exploiting also the spiking times and corresponding previous potential value from the algorithm. Both implementations are provided in the `` `compute_delta_and_I` `` function of the code.

A.1.3. *Maximum Likelihood Estimation.* The MLE $\hat{\theta}_t^N$ is obtained by maximizing the log-likelihood ratio (a discretized version of 5) with respect to $\theta \in \Theta$. This is achieved using the `scipy` function `optimize.minimize` with the L-BFGS-B method, which supports box constraints to enforce $\theta_0 > 0$ and $\theta_2 > 0$. The negative log-likelihood is minimized over the parameter space, with an initial guess of $\theta_{\text{init}} = [1.2, 0.8, 4.0]$ and a maximum of 200 iterations. The bounds for the optimization are set to $(10^{-3}, \infty)$ for θ_0 and θ_2, while θ_1 is unconstrained. This approach is implemented in the `` `estimate_mle_scipy` `` function, which leverages the analytical form of the likelihood ratio to ensure numerical efficiency.

A.1.4. *Statistical Tests.* To assess the empirical distributions of the estimators and LAN quantities, we employ the following multivariate statistical tests, all implemented using the `pingouin` and `scipy.stats` libraries in Python:

⋆ Henze-Zirkler Test: A multivariate normality test based on the work of [26]. This test evaluates whether the empirical distribution of Δ_t^{N,θ^*}, the LAN predictor, or the MLE matches a multivariate normal distribution. The null hypothesis is that the data is normally distributed.

⋆ Hotelling's T^2 Test: A test for the mean vector of a multivariate distribution [28]. For Δ_t^{N,θ^*}, the null hypothesis is that the mean is zero, while for the LAN predictor and MLE, it tests whether the mean matches the true parameter θ^*.

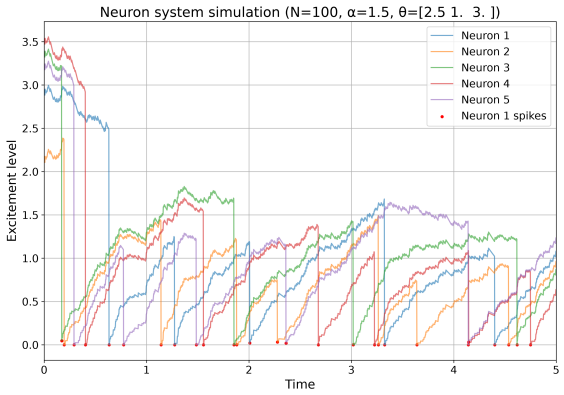
⋆ Mauchly's Test: A test for sphericity, which checks whether the covariance matrix of a dataset is proportional to the identity matrix [32]. In our context, we apply this test to the transformed data (e.g., $(\hat{I}_t^{N,\theta_*})^{1/2}(\hat{\theta}_t^N - \theta^*)$) to verify whether the empirical covariance matches the theoretical covariance structure predicted by the Fisher information matrix.

All tests are performed at a significance level of $\alpha = 0.05$, and the results are reported in the CSV files `test_results_*.csv` for Δ_t^{N,θ^*} (abbreviated as `delta`) and $\hat{\theta}_t^N$ (`MLE`), but also $\widetilde{\theta}_t^N$ (`LAN`).

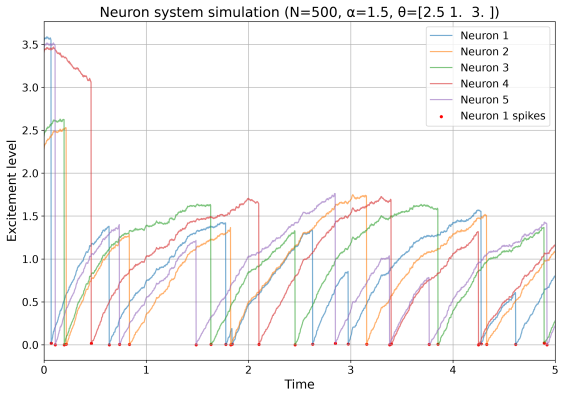
A.2. **Other parameter values.** Alternative scenarios have been considered. First, we will discuss the effect of increasing the number of neurons in Section A.2.1, thus making the system closer to the McKean-Vlasov dynamics. Secondly in Section A.2.2, we will present scenarios for which the randomness is more pronounced on the contrary, firstly with a smaller neuron number, then with another set of parameters.

A.2.1. *Larger neuron number.* When we increase the number of neurons to $N = 100$ and even more with $N = 500$, the fluctuations of the system get visibly averaged out, as we can see in Figures 6 and 7 that are to be compared with respectively Figures 2 and 3. In particular for $N = 500$, the dynamics appear very close to deterministic between spikes, while the global spiking intensity appears to stabilize well after 1s.

Next, we compare the empirical and theoretical covariance matrices for Δ_t^{N,θ^*} and the MLE in Figures 8, 9, to be compared with Figure 4. While the quality of approximation appears not to improve

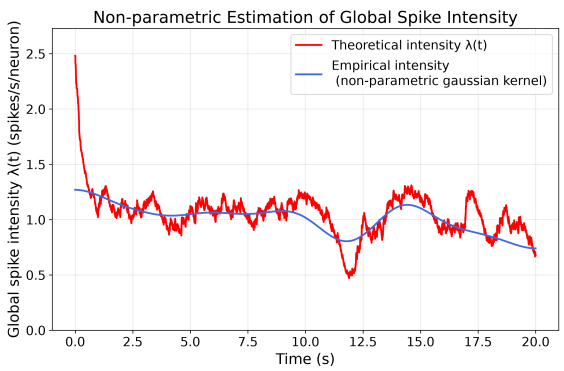


(A) $N = 100$.

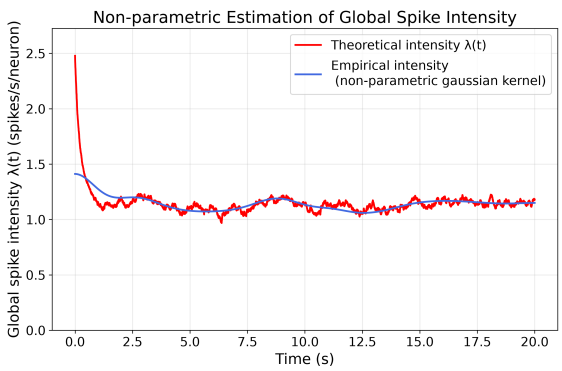


(B) $N = 500$.

FIGURE 6. Illustration of 5 neuron state trajectories over the time-window $[0, 5]$, for $N = 100$ and $N = 500$ neurons.



(A) $N = 100$.



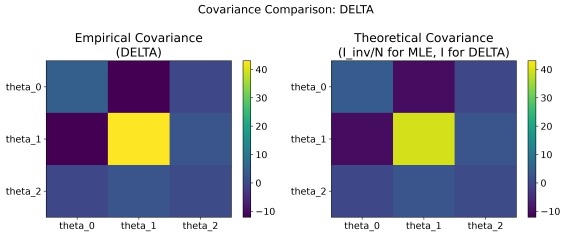
(B) $N = 500$.

FIGURE 7. Estimation of the global spiking intensity for $N = 100$ and $N = 500$ neurons, in blue with a nonparametric gaussian kernel intensity derived from the spike times, in red with the computation of the spiking rate over the neuronal states (for comparison).

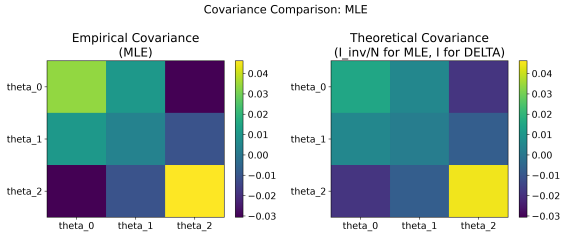
with $N = 100$ as compared to $N = 50$, it is even worse concerning the MLE, an improvement is observed with $N = 500$. Remark that the number of replicates may not be large enough for the empirical covariance matrix to be estimated accurately enough. Note however that these estimations of covariance for the MLE are rescaled by $1/N$, so that the estimator is indeed more efficient as N gets larger.

This inadequacy in the covariance for the replicates where $N = 100$ naturally entails the rejection of Mauchly's test, as we can see in Table 4. In this table, we display the p-values corresponding to our different tests for the varying population sizes under consideration. Most of the other tests are

accepted with large p-values[2]. Nonetheless, we observe that the test of mean value is rejected for both the MLE and Δ_t^{N,θ^*} in the situation where $N = 500$. This outcome is surprising. It may be due to a rare event for the replicates, given that all quantities are affected similarly (in a new set of 100 replicates of the same scenario, this test is far from being rejected). The case where $N = 25$ is displayed as well in Table 4, and will be discussed in the following subsection, because it corresponds to a larger level of randomness.

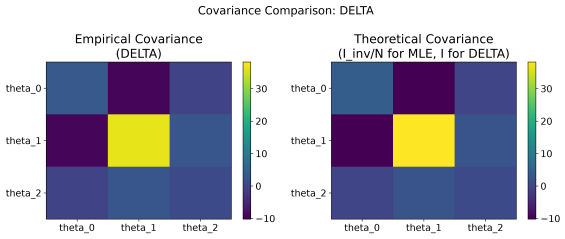


(A) Empirical covariance matrix of Δ_t^{N,θ^*} (over 100 replicates) as compared to $\widehat{I}_t^N$. Relative error in Frobenius norm: 0.10.

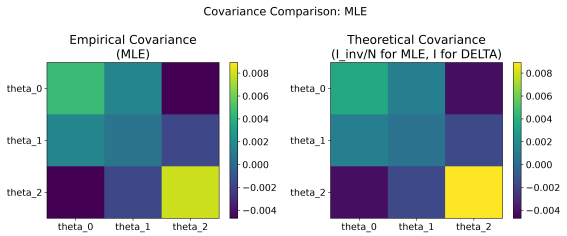


(B) Empirical covariance matrix of the MLE as compared to $N^{-1}(\widehat{I}_t^N)^{-1}$. Relative error in Frobenius norm: 0.48.

FIGURE 8. Comparison between the empirical covariance and the prediction with the estimated Fisher information matrix, when $N = 100$.



(A) Empirical covariance matrix of Δ_t^{N,θ^*} (over 100 replicates) as compared to $\widehat{I}_t^N$. Relative error in Frobenius norm: 0.06.



(B) Empirical covariance matrix of the MLE as compared to $N^{-1}(\widehat{I}_t^N)^{-1}$. Relative error in Frobenius norm: 0.15.

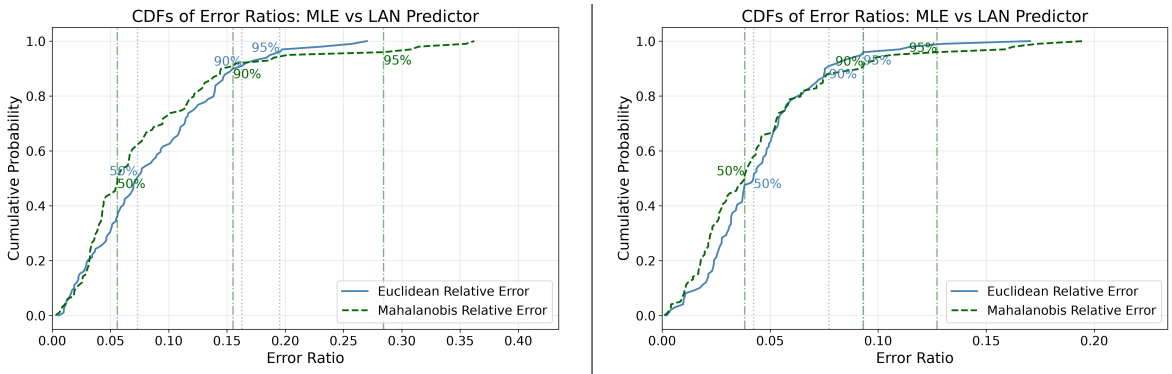
FIGURE 9. Comparison between the empirical covariance and the prediction with the estimated Fisher information matrix, when $N = 500$.

Concerning finally the adequation between the MLE and the LAN predictor, we show in Figure 10 the distribution of relative error that is to be compared with Figure 5-(B). This plot confirms that the approximation gets clearly better as the number of neurons gets larger, even though the MLE gets closer to θ^*. For instance, 90% of the replicates have a relative error below 15% for $N = 100$ and below 10% for $N = 500$ (under both distances in both cases).

[2]While we may expect Δ_t^{N,θ^*} to be closer than the MLE to the expected normal distribution, and the LAN predictor to be in-between, this is not clearly visible from the p-values.

	$N = 25$			$N = 50$			$N = 100$			$N = 500$		
	HZ	HT	M	HZ	HT	M	HZ	HT	M	HZ	HT	M
MLE	0.105	0.161	0.024	0.527	0.518	0.769	0.504	0.072	0.002	0.623	0.016	0.094
LAN	0.077	0.027	0.006	0.746	0.348	0.538	0.420	0.390	0.005	0.442	0.017	0.104
Delta	0.025	0.232	0.050	0.452	0.714	0.205	0.546	0.210	0.996	0.535	0.016	0.918

TABLE 4. Test results given the p-value for different sample sizes (N); methods: HZ, HT and M for respectively Henze-Zirkler, Hotelling's T^2 and Mauchly, that are tests for respectively normality, mean and sphericity; and quantities: MLE, LAN and $Delta$ stands respectively for $\hat{\theta}_t^N$, $\check{\theta}_t^N$ and $\widehat{\Delta}_t^{N,\theta^*}/\sqrt{N}$. Accepted if $p > 0.05$, Rejected (in gray) otherwise.



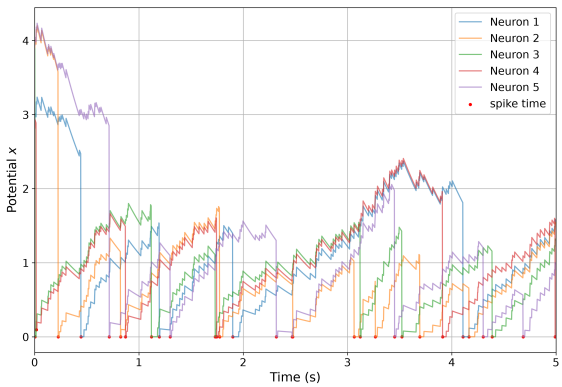
(A) $N = 100$. (B) $N = 500$.

FIGURE 10. Comparison between the estimated MLE $\hat{\theta}_t^N$ and the LAN predictor $\check{\theta}_t^N = \theta^* + (\widehat{I}_t^{N,\theta^*})^{-1}\widehat{\Delta}_t^{N,\theta^*}/\sqrt{N}$ as compared to θ^*, for $N = 100$ and $N = 500$ neurons. C.d.f. of the relative error $\|\hat{\theta}_t^N - \check{\theta}_t^N\|/\|\hat{\theta}_t^N - \theta_*\|$ for the Euclidian distance (blue line) and the Mahalanobis distance (green dashed line). Vertical lines display the quantiles of both distributions corresponding to 50%, 90% and 95%.

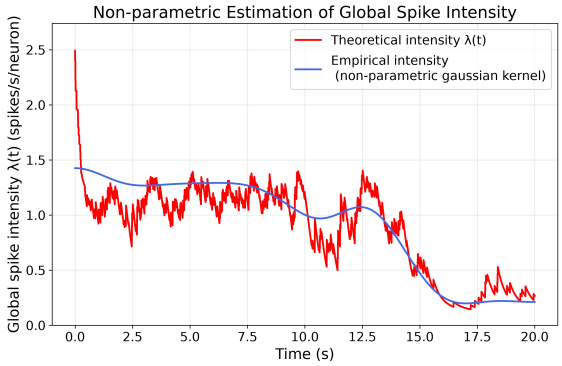
A.2.2. *Larger fluctuation levels.* The effect of a larger level of randomness can first be identified by looking at the scenario with $N = 25$, as it was hinted in the discussion of Table 4. There, we can see that some tests are rejected, especially the ones regarding the covariance matrix, and that the MLE surprisingly scores better for the two other tests.

That the fluctuations of the system do visibly not get averaged out when $N = 25$ can be realized by looking at Figure 11 whose panels are to be compared respectively with Figures 2 and 3. In particular with the estimation of the global spiking intensity in panel (B), we can observe an instance of extinction of the system, after an intermediate period of stabilization.

Next, we look at the covariance matrix of the MLE and the relative error with the LAN predictor in Figure 12, that is to be compared with Figure 4. Despite these large randomness levels, the empirical covariance matrix of the MLE fits the expected matrix quite well. The relative error is undoubtedly larger. Yet more than 50% of the replicates have a relative error below 20%, and more than 95% below 40%, which is still significant.

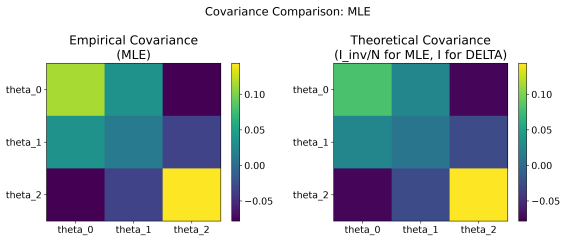


(A) Neuron trajectories.

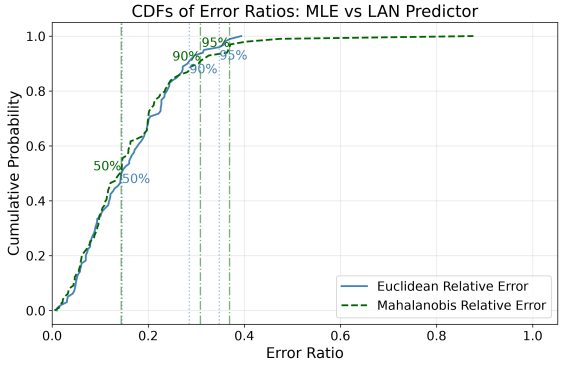


(B) Estimated global spiking intensity. The extinction of the system can be observed.

FIGURE 11. Visualisation of the randomness level for $N = 25$.



(A) Empirical covariance matrix of the MLE as compared to $N^{-1}(\widehat{I}_t^N)^{-1}$. Relative error in Frobenius norm: 0.20.



(B) C.d.f. of the relative error $\|\hat{\theta}_t^N - \check{\theta}_t^N\|/\|\hat{\theta}_t^N - \theta_*\|$.

FIGURE 12. Comparison between the LAN prediction and the MLE, for $N = 25$.

Finally, we describe the outcome of replicates generated with another set of parameters, given in Table 5, and for which the randomness level is amplified with still $N = 50$. As shown in Figure 14-(B), the spiking rate is smaller, and while the theoretical intensity corresponding to neuronal stats stabilizes quickly and remains quite confined, there are large variations in the observed intensity. So neuron trajectories are affected by less frequent oscillations yet more variations in the effective slope, as we can see in Figure 14-(A). The covariance matrices are also quite badly conditioned, in that there is much more precision in θ_0 (as we can see in Figure 15-(A)). and much less in θ_2 (as we can see in Figure 15-(B)). The relative error (for the covariance matrices in Frobenius norm) is actually reduced in these conditions, to 1.4% for Δ_t^{N,θ^*} and to 14% for the MLE, though it is difficult to identify visually.

N	α	ν_1	ν_2	θ_0	θ_1	θ_2
50	0.3	10	$10 = \nu_1$	0.8	0.6	3.0

TABLE 5. Alternative set of parameters.

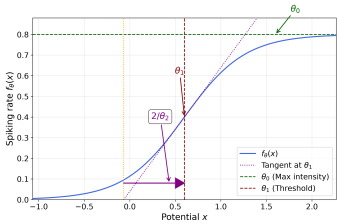
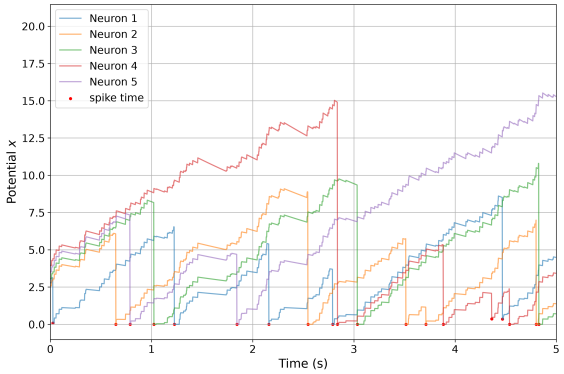
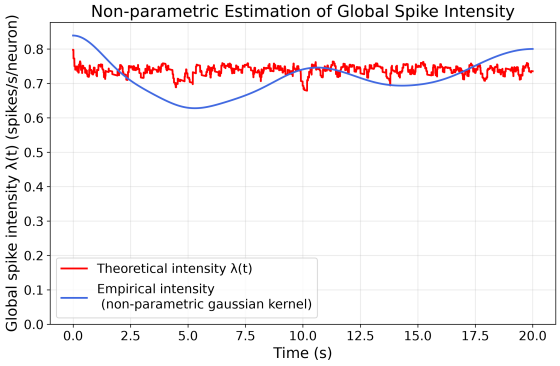


FIGURE 13. Spiking rate, for those new parameters.

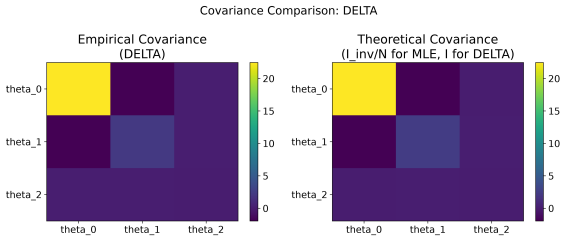


(A) Neuron trajectories.

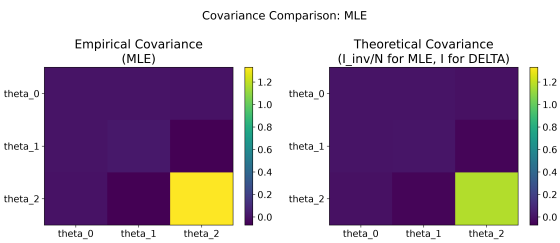


(B) Estimated global spiking intensity. The extinction of the system can be observed.

FIGURE 14. Visualisation of the randomness level, for the set of parameters given in Table 5.



(A) Empirical covariance matrix of Δ_t^{N,θ^*} (over 100 replicates) as compared to $\widehat{I}_t^N$. Relative error in Frobenius norm: 0.014.

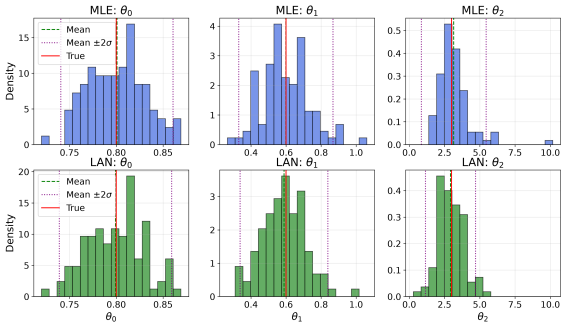


(B) Empirical covariance matrix of the MLE as compared to $N^{-1}(\widehat{I}_t^N)^{-1}$. Relative error in Frobenius norm: 0.14.

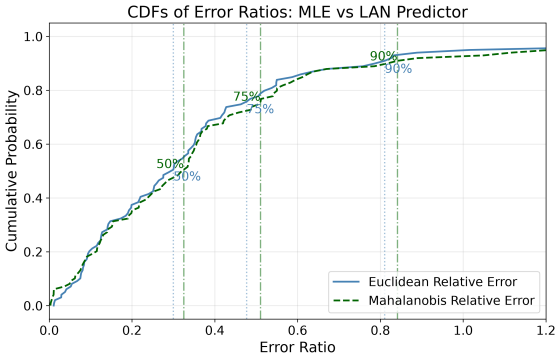
FIGURE 15. Comparison between the empirical covariance and the prediction with the estimated Fisher information matrix, for the set of parameters given in Table 5.

Table 6 displays the p-values corresponding to the different tests we conduct as well for this set of parameters. As we can see, the test of sphericity is rejected for the MLE (as well as the LAN predictor, and contrary to Δ_t^{N,θ^*}). We suspect it is typically the one most susceptible to be rejected, especially

for a case where the randomness in the Fisher information could not be neglected, notably when it is badly conditioned. The compared histograms and cumulative distribution function of the relative error between the MLE and the LAN predictor are also provided in Figure 16. It is noticeable from the later that the deviations are much worse than for our reference set of parameters, see Figure 16.



(A) Histograms of the MLE (top row) and LAN predictor (bottom row) for the three parameters θ_0, θ_1, and θ_2.



(B) C.d.f. of the relative error $\|\hat{\theta}_t^N - \check{\theta}_t^N\|/\|\hat{\theta}_t^N - \theta_*\|$.

FIGURE 16. Comparison between the estimated MLE $\hat{\theta}_t^N$ and the LAN predictor, for the set of parameters given in Table 5.

	HZ	HT	M
MLE	$3.2\,10^{-4}$	0.54	$8.5\,10^{-3}$
LAN	0.60	0.38	$5.6\,10^{-3}$
Delta	0.58	0.36	0.71

TABLE 6. Test results given the p-value for the parameter set given in Table 5; methods: HZ, HT and M for respectively Henze-Zirkler, Hotelling's T^2 and Mauchly, that are tests for respectively normality, mean and sphericity; and quantities: MLE, LAN and $Delta$ stands respectively for $\hat{\theta}_t^N$, $\check{\theta}_t^N$ and $\widehat{\Delta}_t^{N,\theta^*}/\sqrt{N}$. Accepted if $p > 0.05$, Rejected (in gray) otherwise.

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